

# Central-Charge Recursion for Genus- $g$ , $N$ -Point Neveu–Schwarz Superconformal Blocks

Pole factorization, the global  $\mathfrak{osp}(1|2)$  network, and the vacuum seed

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## Abstract

We derive the central-charge recursion for an all-Neveu–Schwarz chiral block on a genus- $g$ ,  $N$ -punctured surface in the plumbing frame of Cho–Collier–Yin. At fixed plumbing degree the block is meromorphic in the ordinary central charge. Its poles occur when an internal NS Verma module develops a singular vector. The residue is local to that edge. For distinct endpoints it is the product of the pole Jacobian, inverse null-norm slope, and one ordered NS fusion polynomial at each adjacent trinion; a self-loop instead uses an incidence-ordered toric factor with a shifted second argument and an intermediate sector toggle.

The regular term is derived directly in components. At large  $c$ , the module separates into a universal non-global vacuum Fock space and a global  $\mathfrak{osp}(1|2)$  module. We give the complete component three-point kernel, the global graph sum, the lifted Schottky product for the vacuum factor, and the Koszul sign that assembles the two factors. We also give an explicit sign compiler: the ordered endpoint phase follows from the graded slot permutation and the NS reflection identity, while the linear BPZ/spin-frame bits follow from the lifted odd local coordinates and sewing transitions. These data give an all-level expression for the genus- $g$ ,  $N$ -point regular block.

After the component derivation is complete, we record two optional reformulations. The first packages the global trinion and its edge-parity contractions in superspace. The second introduces the genuine odd moduli of an NS pants decomposition. The latter is kept as an exploratory local plumbing description and is not used in the derivation of the recursion. Direct PBW sewing at genus one and genus two provides independent coefficient checks through the stated level cutoffs. The recursion theorem is for generic internal weights. Coincident Kac loci can produce genuine higher-order poles and require an additional detuning or Laurent-jet prescription that is not developed here.

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## Introduction and statement of the result

Let  $\Gamma$  be a trivalent pants graph for a stable genus- $g$  surface with  $N$  NS punctures. It has

$$|E_{\text{int}}(\Gamma)| = 3g - 3 + N, \quad |V(\Gamma)| = 2g - 2 + N. \quad (1)$$

An internal edge  $e$  carries an NS weight  $h_e$ , a reduced plumbing parameter  $q_e$ , and a choice of square-root lift. A vertex  $v$  carries one of the two NS three-point structures  $\alpha_v \in \{0, 1\}$ . Primary propagation factors  $\prod_e q_e^{h_e}$  are stripped off.

For generic weights, the reduced block obeys

$$\begin{aligned} \mathcal{F}_\Gamma^\alpha(c, \mathbf{h}; \mathbf{q}, \boldsymbol{\xi}) &= \mathcal{U}_\Gamma^\alpha(\mathbf{h}; \mathbf{q}, \boldsymbol{\xi}) \\ &+ \sum_{e \in E_{\text{int}}(\Gamma)} \sum_{\substack{r \geq 2, \ s \geq 1 \\ r+s \in 2\mathbb{Z}}} \frac{q_e^{rs/2} \widehat{\mathcal{R}}_{e;r,s}(\mathbf{h})}{c - c_{r,s}(h_e)} \\ &\times \mathcal{K}_{\Gamma,e}^{\pi_{r,s}} \mathcal{F}_\Gamma^{\mathbb{T}_e^{\pi_{r,s}}} \alpha \left( c_{r,s}(h_e), \mathbf{h} + \frac{rs}{2} \mathbf{e}_e; \mathbf{q}, \boldsymbol{\xi} \right). \end{aligned} \quad (2)$$

Here  $\mathbb{T}_e^p$  toggles the three-form label once per incidence (half-edge) of  $e$ . It therefore toggles both endpoint labels for an ordinary edge with distinct endpoints, while a self-loop toggles the same vertex twice and hence acts as the identity. At a self-loop the intermediate label is nevertheless toggled between the two sequential null factorizations, as made explicit in Eq. (45). The diagonal Koszul transport  $\mathcal{K}_{\Gamma,e}$  is computed from the lifted plumbing frames; its coefficient definition is given below and is nontrivial only for an odd null. The lift factor at the null level is contained in the chosen meaning of  $q_e^{rs/2}$ . Equivalently it can be displayed as  $q_e^{\lfloor rs/2 \rfloor} \zeta_e^{rs \bmod 2} q_e^{(rs \bmod 2)/2}$ . The local residue action is

$$\widehat{\mathcal{R}}_{e;r,s} = J_{r,s}(h_e) A_{r,s}(h_e) \begin{cases} \widehat{\Sigma}_{r,s}^{(v_L,e)} \widehat{\Sigma}_{r,s}^{(v_R,e)}, & v_L \neq v_R, \\ \widehat{\mathcal{M}}_{r,s}^{(v,e)}, & v_L = v_R = v. \end{cases} \quad (3)$$

The hats matter. The incidence factors are diagonal in the component-parity basis, while  $\widehat{\mathcal{M}}$  is the incidence-ordered self-loop factor. Both are given explicitly below. The regular term is defined by

$$\mathcal{U}_\Gamma^\alpha(\mathbf{h}; \mathbf{q}, \boldsymbol{\xi}) := \lim_{c \rightarrow \infty} \mathcal{F}_\Gamma^\alpha(c, \mathbf{h}; \mathbf{q}, \boldsymbol{\xi}). \quad (4)$$

For primary or global external states this note computes it directly for the genus-two theta graph through total level six and gives its all-order formula for an arbitrary ordered plumbing graph in Eq. (63) below, conditional on a specified ordered set of lifted local plumbing maps.

There are three distinct claims in Eq. (2).

1. The pole position, null norm, fusion factors, and shifted module are established NS representation theory [1, 2].
2. Edge locality and graph assembly follow by the same inverse-Gram argument as CCY [3]. Distinct-endpoint edges factor independently; self-loops use the sequential toric factorization.
3. The component  $\mathfrak{osp}(1|2)$  network gives the global subsector of  $\mathcal{U}_\Gamma$ . The non-global sector resums to the lifted primitive Schottky product, and the polarization of the half-edge permutation gives the explicit Koszul sign that assembles the two sectors. The linear orientation ledger is computed from the odd-coordinate frames and the residual signs in the lifted sewing maps.

Thus Eq. (2), together with Eq. (63), is a higher-genus Zamolodchikov recursion formula at generic weights in a fully specified all-NS plumbing frame, for primary or global external states and with the self-loop rule (45). Extensions outside the present result include higher-order poles on exceptional weight loci and non-global external descendants with their additional large- $c$  scaling.

## 1 Definition of the NS plumbing block

### 1.1 NS algebra and central-charge convention

We use the ordinary central charge, not  $\hat{c} = 2c/3$ :

$$\begin{aligned} [L_m, L_n] &= (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}, \\ [L_m, G_r] &= \left(\frac{m}{2} - r\right)G_{m+r}, \\ \{G_r, G_s\} &= 2L_{r+s} + \frac{c}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0}, \quad r, s \in \mathbb{Z} + \frac{1}{2}. \end{aligned} \quad (5)$$

The super-Liouville parameterization used throughout the repository is

$$Q = b + b^{-1}, \quad c = \frac{3}{2} + 3Q^2 = \frac{3}{2}(1 + 2Q^2), \quad h(\lambda) = \frac{Q^2 - \lambda^2}{8}. \quad (6)$$

At  $b = 1$ , this gives  $c = 27/2$ . Belavin–Geiko denote  $\hat{c} = 2c/3 = 1 + 2Q^2$  by  $c$ . Their pole denominator and Jacobian must be converted together; the ratio  $J/(c - c_*)$  is unchanged.

### 1.2 NS modules, descendants, and three-forms

An NS highest-weight state satisfies

$$L_0 \nu_h = h \nu_h, \quad L_{n>0} \nu_h = G_{r>0} \nu_h = 0, \quad \langle \nu_h | \nu_h \rangle = 1. \quad (7)$$

PBW descendants are

$$G_{-r_1} \cdots G_{-r_p} L_{-n_1} \cdots L_{-n_k} \nu_h, \quad |I| = \sum_a r_a + \sum_b n_b \in \frac{1}{2}\mathbb{Z}_{\geq 0}, \quad (8)$$

with distinct, ordered fermionic modes. Their level- $\ell$  Gram matrix is

$$B_{h,c}^{(\ell)}[I, J] = \langle \nu_h | \mathcal{L}_I^\dagger \mathcal{L}_J | \nu_h \rangle. \quad (9)$$

There are two normalized NS chiral three-forms. We label them by  $\alpha = 0, 1$ , with  $\alpha = 0$  even and  $\alpha = 1$  odd. It is useful to regard

$$\rho^0(\nu_3, \nu_2, \nu_1) = 1, \quad \rho^1(\nu_3, G_{-1/2}\nu_2, \nu_1) = 1 \quad (10)$$

as kinematic normalizations. Physical even and odd structure constants are not part of the block. Fermionic slot order is part of the definition of  $\rho^\alpha$ ; cyclic reordering can produce a Koszul sign.

### 1.3 Plumbing coordinates and the NS spin lift

CCY build a surface from three-holed spheres represented as two-holed discs and use only global Möbius maps for the boundary identifications. The same plane three-forms at  $0, 1, \infty$  are used here. For an NS super-plumbing fixture, local supercoordinates may be written as

$$z' = \frac{q}{z}, \quad \theta' = \pm i \frac{\sqrt{q}}{z} \theta. \quad (11)$$

This is the convention reviewed in the string notes [4]. The sign and the branch of  $\sqrt{q}$  are spin-lift data. We store them as a lifted variable

$$\mathbf{q}^{\text{lift}} = (q_e, \xi_e)_{e \in E_{\text{int}}}, \quad \xi_e \in \{+1, -1\}, \quad (12)$$

The NS label already fixes antiperiodic fermions around the circumference of the tube. The sign  $\xi_e$  does not turn this circle into a Ramond circle. It is the  $\mathbb{Z}_2$  transition function that transports the spin bundle *through* the tube. In operator language the lifted propagation is

$$\mathbf{P}_e(q_e, \xi_e) = q_e^{L_0 - h_e} \xi_e^F, \quad |I| = n + \frac{\epsilon}{2}, \quad \mathbf{P}_e \mathcal{L}_I \nu_{h_e} = q_e^{n + \epsilon/2} \xi_e^\epsilon \mathcal{L}_I \nu_{h_e}, \quad (13)$$

where  $F$  is fermion parity and  $\epsilon = 0, 1$ . Equivalently one may choose the lifted coordinate  $x_e = \xi_e \sqrt{q_e}$  and write the weight as  $q_e^n x_e^\epsilon$ .

An individual edge sign depends on the odd-coordinate convention on the adjacent pairs of pants. Redefining such a coordinate can change the representative signs on adjacent edges. The invariant data are the products of the lifted transition functions around closed graph cycles; these products give the global spin holonomies. For example, closing one NS tube into a torus gives

$$\text{Tr}_{\mathcal{H}_{\text{NS}}} \left( \xi^F q^{L_0 - c/24} \right), \quad (14)$$

so  $\xi = +1$  is the ordinary NS trace and  $\xi = -1$  inserts  $(-1)^F$ . Thus the NS module fixes the spin structure around one torus cycle, while  $\xi$  distinguishes the spin holonomy around the cycle created by closing the tube.

The harmless fixed phase  $i$  in Eq. (11) can be absorbed into the fermionic BPZ sewing map; it must be chosen once, not independently at each coefficient.

## 1.4 State-sewing definition

Choose an orientation of every internal edge. Let  $I_e$  and  $J_e$  label equal-level descendants at its two ends. At each external puncture  $i = 1, \dots, N$ , fix a primary or global  $\mathfrak{osp}(1|2)$  descendant

$$|\Psi_i\rangle = L_{-1}^{n_i} G_{-1/2}^{\eta_i} \nu_{d_i}, \quad L_0 |\Psi_i\rangle = h_i^{\text{ext}} |\Psi_i\rangle, \quad h_i^{\text{ext}} = d_i + n_i + \frac{\eta_i}{2}, \quad n_i \in \mathbb{Z}_{\geq 0}, \quad \eta_i \in \{0, 1\}. \quad (15)$$

Thus a bottom component has  $(n_i, \eta_i) = (0, 0)$ , while a top component has  $(n_i, \eta_i) = (0, 1)$ . Non-global external descendants are not included in the theorem. This restriction is necessary: for example

$$\rho(L_{-2}\nu_h, \mathbf{1}, L_{-2}\nu_h) = \langle \nu_h | L_2 L_{-2} | \nu_h \rangle = 4h + \frac{c}{2}, \quad (16)$$

which has no internal pole but grows with  $c$ . Such insertions require external vacuum factors and an amended entire part. We suppress the permitted fixed external data in the arguments of the block.

With the primary propagation factors  $\prod_e q_e^{h_e}$  removed, we define the reduced genus- $g$ ,  $N$ -point NS super-Virasoro block associated with the ordered plumbing graph  $\Gamma$  by

$$\begin{aligned} \mathcal{F}_\Gamma^\alpha(c, h; \mathbf{q}, \xi) = & \sum_{\ell \in (\frac{1}{2}\mathbb{Z}_{\geq 0})^{E_{\text{int}}}} \prod_e q_e^{\ell_e} \xi_e^{2\ell_e} \sum_{\substack{|I_e|=|J_e|=\ell_e \\ e \in E_{\text{int}}}} \Omega_\Gamma(\mathbf{I}, \mathbf{J}; \boldsymbol{\eta}) \left[ \prod_e (B_{h_e, c}^{(\ell_e)})^{-1} [I_e, J_e] \right] \\ & \times \left[ \prod_{v \in V(\Gamma)} \rho_v^{\alpha_v} \left( \{ \mathcal{L}_{K(v, e)} \nu_{h_e} \}_{e \in E_{\text{int}}(v)}; \{ |\Psi_i\rangle \}_{i \in E_{\text{ext}}(v)} \right) \right]. \quad (17) \end{aligned}$$

Since  $\xi_e^2 = 1$ , the factor  $\xi_e^{2\ell_e}$  is one at integral level and  $\xi_e$  at half-integral level. When a tube is closed into a handle,  $\xi_e = \pm 1$  distinguishes the two spin holonomies along the resulting torus cycle; on a general plumbing graph the invariant spin data are the corresponding products around closed cycles. Here  $E_{\text{int}}(v)$  and  $E_{\text{ext}}(v)$  are the internal and external half-edges incident on  $v$ . The label  $K(v, e)$  is  $I_e$  or  $J_e$  according to the edge orientation and the canonical order of the three half-edges at  $v$ .

The factor  $\Omega_\Gamma(\mathbf{I}, \mathbf{J}; \boldsymbol{\eta})$  is the global graded sewing sign obtained by permuting all parity-homogeneous half-edge states from vertex slot order to contraction order, including the fixed external parities  $\boldsymbol{\eta}$ . In the chosen lifted plumbing frame it is  $(-1)^{Q_\Gamma}$ , with the explicit orientation character  $Q_\Gamma$  defined in Eq. (60). This factor is part of the finite- $c$  definition, not only of its large- $c$  limit.

This definition contains no cylinder Casimir factor  $q^{-c/24}$ , no superghost determinant, no supermoduli measure, and no structure constants. A change of local-coordinate frame can multiply it by a conformal-anomaly factor. Such a block must not be inserted into the recursion without first translating back to the CCY frame.

## 2 Meromorphic ansatz and representation-theoretic residues

### 2.1 Partial-fraction ansatz

At fixed internal weights, the block is written as

$$\mathcal{F}_\Gamma(c) = \mathcal{U}_\Gamma + \sum_{e \in E_{\text{int}}(\Gamma)} \sum_{\substack{r \geq 2, \ s \geq 1 \\ r+s \in 2\mathbb{Z}}} \frac{\text{Res}_{c=c_{r,s}(h_e)} \mathcal{F}_\Gamma}{c - c_{r,s}(h_e)}. \quad (18)$$

The remainder of this section determines the poles and residues from NS representation theory. Section 3 computes the entire part  $\mathcal{U}_\Gamma$ .

Fix a multi-level  $\ell$ . Gram entries and descendant three-forms are polynomial in  $c$  and the weights. Their contraction is therefore a rational function of  $c$ . At generic weights its poles are the Kac zeros of one internal Gram matrix. At fixed  $\ell$ , only finitely many  $(r, s)$  obey  $rs/2 \leq \ell_e$ , so the partial-fraction decomposition is finite. For the primary/global external states in Eq. (15), the coefficient remains finite at  $c = \infty$ , so its entire part is a constant equal to that limit. Equation (16) shows why this conclusion does not extend unchanged to arbitrary external descendants.

## 2.2 NS Kac poles on the $c$ -plane

### 2.2.1 Degenerate weight and pole branch

An NS Verma module is degenerate for

$$r, s \in \mathbb{Z}_{>0}, \quad r + s \in 2\mathbb{Z}, \quad \lambda_{r,s} = rb + sb^{-1}, \quad (19)$$

at the weight

$$h_{r,s}(c) = \frac{(b + b^{-1})^2 - (rb + sb^{-1})^2}{8}. \quad (20)$$

The singular vector has level and parity

$$N_{r,s} = \frac{rs}{2}, \quad \pi_{r,s} = rs \pmod{2}. \quad (21)$$

For  $c$ -recursion, the  $(1, 1)$  zero does not move with  $c$ . We use  $r \geq 2, s \geq 1$  to choose one representative from the  $b \leftrightarrow b^{-1}$  pair.

At fixed internal weight  $h$ , set  $x = b^2$ . The pole equation is

$$(r^2 - 1)x^2 + [8h + 2(rs - 1)]x + (s^2 - 1) = 0. \quad (22)$$

The branch used by the project code is

$$D_{r,s}(h) = \sqrt{16h^2 + 8(rs - 1)h + (r - s)^2}, \quad (23)$$

$$x_{r,s}(h) = -\frac{4h + rs - 1 + D_{r,s}(h)}{r^2 - 1}, \quad b_{r,s}(h) = \sqrt{x_{r,s}(h)},$$

$$c_{r,s}(h) = \frac{15}{2} + 3x_{r,s}(h) + \frac{3}{x_{r,s}(h)}. \quad (24)$$

Square roots are chosen once and analytically continued. Reselecting a principal branch inside a recursion tree can jump between poles.

### 2.2.2 Pole Jacobian

Define

$$J_{r,s}(h) = -\frac{\partial c_{r,s}(h)}{\partial h}. \quad (25)$$

For direct implementation,

$$\frac{\partial x_{r,s}}{\partial h} = -\frac{1}{r^2 - 1} \left[ 4 + \frac{16h + 4(rs - 1)}{D_{r,s}(h)} \right],$$

$$J_{r,s}(h) = -3 \left(1 - x_{r,s}^{-2}\right) \frac{\partial x_{r,s}}{\partial h}. \quad (26)$$

The sign follows from expanding the null norm at fixed  $h$ :

$$h - h_{r,s}(c) = - \left. \frac{\partial h_{r,s}}{\partial c} \right|_{c_*} (c - c_*) + O((c - c_*)^2), \quad c_* = c_{r,s}(h). \quad (27)$$

## 2.3 Null norm and ordered fusion factors

### 2.3.1 Singular-vector normalization

Normalize the singular operator by its global leading monomial,

$$\chi_{r,s} = D_{r,s} \nu_{h_{r,s}}, \quad D_{r,s} = \begin{cases} L_{-1}^{rs/2} + \dots, & rs \text{ even}, \\ G_{-1/2} L_{-1}^{(rs-1)/2} + \dots, & rs \text{ odd}. \end{cases} \quad (28)$$

Equivalently the coefficient of  $G_{-1/2}^{rs}$  is one. Near the Kac zero, its norm is

$$\langle \chi_{r,s}(h) | \chi_{r,s}(h) \rangle = \frac{h - h_{r,s}(c)}{A_{r,s}(c)} + O((h - h_{r,s})^2). \quad (29)$$

In the convention of Refs. [1, 2],

$$A_{r,s}(h) = \frac{1}{2} \prod_{\substack{p=1-r,\dots,r \\ q=1-s,\dots,s \\ p+q \in 2\mathbb{Z} \\ (p,q) \neq (0,0),(r,s)}} \left[ \frac{pb_{r,s} + qb_{r,s}^{-1}}{\sqrt{2}} \right]^{-1}. \quad (30)$$

Here and below  $b_{r,s} = b_{r,s}(h)$  is evaluated at the pole.

### 2.3.2 Two NS fusion polynomials

For an adjacent leg of weight  $d_i$ , choose a continuous branch

$$d_i = \frac{(b_{r,s} + b_{r,s}^{-1})^2 - \lambda_i^2}{8}. \quad (31)$$

The two NS fusion polynomials are

$$P_{r,s}^\alpha(d_i, d_j; h) = \prod_{\substack{p=1-r, 3-r, \dots, r-1 \\ q=1-s, 3-s, \dots, s-1 \\ p+q-r-s \in \mathcal{C}_\alpha}} \left( \frac{\lambda_i - \lambda_j + pb_{r,s} + qb_{r,s}^{-1}}{2\sqrt{2}} \right) \times \left( \frac{\lambda_i + \lambda_j + pb_{r,s} + qb_{r,s}^{-1}}{2\sqrt{2}} \right),$$

where

$$\mathcal{C}_0 = 4\mathbb{Z} + 2, \quad \mathcal{C}_1 = 4\mathbb{Z}. \quad (32)$$

The label  $\alpha$  in  $P_{r,s}^\alpha$  is the label of the *original* three-form. If the singular vector is odd, it is the three-form label of the shifted tensor that changes:

$$\alpha' = \alpha \oplus \pi_{r,s}, \quad \pi_{r,s} = rs \pmod{2}. \quad (33)$$

Thus one must not replace  $P_{r,s}^\alpha$  by  $P_{r,s}^{1-\alpha}$  for odd  $rs$ . The direct  $(3,1)$  residue at level  $3/2$  distinguishes these two prescriptions already at one edge.



### 2.3.3 Orientation ledger

The polynomial alone is not the full endpoint factor, but its sign can be computed without a separate table. Fix the canonical trinion slot order  $(\infty, 1, 0)$ . At an incidence  $(v, e)$ , let  $\mathbf{p}_v = (p_\infty, p_1, p_0)$  be the component parities and let  $\pi_{v,e}$  list the old slots in the new order which places the null leg in canonical slot  $a = 1$  or  $a = 3$ . Define

$$\kappa(\pi_{v,e}; \mathbf{p}_v) := \sum_{\substack{i < j \\ \pi_{v,e}^{-1}(i) > \pi_{v,e}^{-1}(j)}} p_i p_j \pmod{2}, \quad (34)$$

the Koszul parity of that literal slot permutation. In the resulting canonical order write  $b = \tilde{p}_2$  for the parity of the middle component and  $p = rs \pmod{2}$  for the parity of the singular vector.

The reflection identities of Refs. [1, 2] then give

$$\rho^\alpha(\chi_{r,s}, \nu_i, \nu_j) = P_{r,s}^\alpha(d_i, d_j; h), \quad (35)$$

$$\rho^\alpha(\nu_i, \nu_j, \chi_{r,s}) = (-1)^{pb} P_{r,s}^\alpha(d_j, d_i; h). \quad (36)$$

Here  $P^\alpha$  uses the fixed-parity convention of Eq. (33): the label is that of the original three-form, while an odd null toggles the shifted tensor. In the component notation of Ref. [2],  $b = 0, 1$  is precisely the unstarred/starred canonical middle component, so Eq. (36) is their  $S_n^0 = 1$ ,  $S_n^1 = (-1)^n$ .

Consequently the complete *canonical endpoint sign* is

$$s_{\text{can}}(v, e; r, s) = (-1)^{\kappa(\pi_{v,e}; \mathbf{p}_v) + pb} \mathbf{1}_{\{a=3\}}, \quad (37)$$

and the endpoint operator is diagonal in the resolved component-parity basis, with eigenvalue

$$\widehat{\Sigma}_{r,s}^{(v,e)} \Big|_{\mathbf{p}_v} = s_{\text{can}}(v, e; r, s) P_{r,s}^{\alpha_v}(d_{v,e}^{(1)}, d_{v,e}^{(2)}; h_e). \quad (38)$$

This canonicalization must happen before the left and right factors are multiplied. For a general multi-edge trinion the entries of  $\mathbf{p}_v$  are summed variables, so  $\widehat{\Sigma}$  cannot be replaced by a scalar depending only on weights. A topology formula with an undifferentiated symbol  $P_{r,s}^2$  loses both this diagonal action and the slot-dependent signs.

There is one important bookkeeping rule. If the odd coordinate on half-edge  $h = (v, e)$  differs from its canonical trinion frame by  $\theta_h = (-1)^{f_h} \theta_h^{\text{can}}$ , then an odd null would contribute the additional incidence factor  $(-1)^{pf_h}$ . In the master recursion we do *not* put this factor into  $\widehat{\Sigma}$ , because it is already included once in  $Q_\Gamma$  and hence in the transport operator  $\mathcal{K}_{\Gamma,e}$ , as derived below. If one instead uses unoriented scalar graph coefficients, the combined incidence sign is

$$s_{\text{inc}}(v, e; r, s) = s_{\text{can}}(v, e; r, s) (-1)^{pf_h}. \quad (39)$$

For example, on the BPZ-reflected trinion of the theta chart the frame bits are  $(f_\infty, f_1, f_0) = (1, 0, 0)$ . Thus the odd  $(3, 1)$  null has combined signs  $(-1, +1, +1)$  in slots  $(\infty, 1, 0)$ , while its canonical endpoint signs for even neighboring primaries are all  $+1$ . This distinction prevents the theta frame sign from being counted twice.

## 2.4 Edge-local residue

At  $c = c_* = c_{r,s}(h_e)$ , choose a basis at levels  $N_{r,s} + n$  that contains all states  $\mathcal{L}_K \chi_{r,s}$ . Only this null-submodule block can be singular. The singular part of the inverse Gram matrix is

$$\text{Res}_{c=c_*} \left( B_{h_e, c}^{(N_{r,s}+n)} \right)^{-1} [\mathcal{L}_K \chi_{r,s}, \mathcal{L}_L \chi_{r,s}] = J_{r,s}(h_e) A_{r,s}(h_e) \left( B_{h_e + N_{r,s}, c_*}^{(n)} \right)^{-1} [K, L]. \quad (40)$$

The shifted weight appears because the singular vector is itself a highest weight state of weight

$$h'_e = h_e + N_{r,s} = h_e + \frac{rs}{2}. \quad (41)$$

The Ward identities factor every descendant three-form above the null state. At one resolved incidence,

$$\rho_v^\alpha(\dots, \mathcal{L}_K \chi_{r,s}, \dots) = \widehat{\Sigma}_{r,s}^{(v,e)} \Big|_{\mathbf{p}_v} \rho_v^{\alpha \oplus \pi_{r,s}}(\dots, \mathcal{L}_K \nu_{h'_e}, \dots). \quad (42)$$

For an ordinary edge  $v_L \neq v_R$ , the two incidences belong to independent three-forms and Eq. (42) may be applied once at each endpoint. All other edges and vertices are spectators. The level carried by the null state supplies  $q_e^{N_{r,s}}$ , giving

$$\text{Res}_{c=c_{r,s}(h_e)} \mathcal{F}_\Gamma^\alpha = q_e^{rs/2} J_{r,s} A_{r,s} \widehat{\Sigma}_{r,s}^{(v_L,e)} \widehat{\Sigma}_{r,s}^{(v_R,e)} \mathcal{K}_{\Gamma,e}^{\pi_{r,s}} \mathcal{F}_\Gamma^{\pi_{r,s}} \alpha \left( c_{r,s}(h_e), \mathbf{h} + \frac{rs}{2} \mathbf{e}_e \right), \quad v_L \neq v_R. \quad (43)$$

If  $e$  is a self-loop, the two null states occur in the same matrix element,

$$\rho^{\alpha_v}(\mathcal{L}_K \chi_{r,s}, \nu_{d_v}, \mathcal{L}_L \chi_{r,s}), \quad (44)$$

and the two factorizations are not independent. Order the two incidences  $a_1, a_2$  by the canonical trinion slots. Factoring  $a_1$  first changes the other sewn weight from  $h_e$  to  $h'_e = h_e + N_{r,s}$  and changes the intermediate three-form label to  $\alpha_v \oplus \pi_{r,s}$ . Therefore the diagonal eigenvalue of the incidence-ordered operator is

$$\widehat{\mathcal{M}}_{r,s}^{(v,e)} \Big|_{\mathbf{p}_v} = s_{\text{loop}}(v, e; r, s; \mathbf{p}_v) P_{r,s}^{\alpha_v \oplus \pi_{r,s}}(d_v, h_e + N_{r,s}; h_e) P_{r,s}^{\alpha_v}(d_v, h_e; h_e). \quad (45)$$

Here  $s_{\text{loop}}$  is obtained by applying the canonical incidence sign (37) first in the original tensor and then in the intermediate tensor, including its toggled parity. In the standard torus rephasing it is the usual toric sign  $S_{r,s}^{\alpha_v}$ . Equation (45) is the toric null matrix element of Ref. [2]; notably, one polynomial sees the shifted weight and, for an odd null, the toggled intermediate sector. The self-loop residue is consequently

$$\text{Res}_{c=c_{r,s}(h_e)} \mathcal{F}_\Gamma^\alpha = q_e^{rs/2} J_{r,s} A_{r,s} \widehat{\mathcal{M}}_{r,s}^{(v,e)} \mathcal{K}_{\Gamma,e}^{\pi_{r,s}} \mathcal{F}_\Gamma^\alpha \left( c_{r,s}(h_e), \mathbf{h} + \frac{rs}{2} \mathbf{e}_e \right), \quad v_L = v_R = v. \quad (46)$$

The final label is unchanged because it is toggled twice. Equations (43) and (46) prove the polar term in Eq. (2). The operation remains edge-local, but a self-loop is local to one trinion and must retain its incidence order.

## 2.5 Complete recursion

### 2.5.1 Functional form

Once the regular term of Section 3 is supplied, substituting it and the local residues (43) and (46) into the partial-fraction decomposition gives the master formula (2). For  $v_L \neq v_R$ , the componentwise eigenvalue of the residue action, written with all local arguments, is

$$\widehat{\mathcal{R}}_{e;r,s}(\mathbf{h}) \Big|_{\mathbf{p}} = J_{r,s}(h_e) A_{r,s}(h_e) \widehat{\Sigma}_{r,s}^{(v_L,e)} \Big|_{\mathbf{p}_{v_L}} (h_{L,1}, h_{L,2}; h_e, \alpha_{v_L}) \widehat{\Sigma}_{r,s}^{(v_R,e)} \Big|_{\mathbf{p}_{v_R}} (h_{R,1}, h_{R,2}; h_e, \alpha_{v_R}), \quad (47)$$

where  $h_{L,a}$  and  $h_{R,a}$  are the other two weights at the adjacent vertices. They can be internal or external weights. The canonical slot rotation decides whether each  $\widehat{\Sigma}$  uses Eq. (35) or (36). For a self-loop, Eq. (45) replaces Eq. (47).

### 2.5.2 Triangular coefficient form

Expand

$$\mathcal{F}_\Gamma^\alpha = \sum_{\ell} F_\ell^\alpha(c, \mathbf{h}; \boldsymbol{\xi}) \prod_e q_e^{\ell_e}, \quad \mathcal{U}_\Gamma^\alpha = \sum_{\ell} U_\ell^\alpha(\mathbf{h}; \boldsymbol{\xi}) \prod_e q_e^{\ell_e}. \quad (48)$$

For a multi-level  $\ell$ , write  $\epsilon_e(\ell) = 2\ell_e \pmod{2}$ . The coefficient action of the Koszul transport is

$$\kappa_{\Gamma,e}^{(p)}(\ell) := \frac{\Omega_\Gamma(\boldsymbol{\epsilon}(\ell))}{\Omega_\Gamma(\boldsymbol{\epsilon}(\ell - p\mathbf{e}_e/2))}, \quad p \in \{0, 1\}. \quad (49)$$

It is one for  $p = 0$ . This ratio is the coefficientwise meaning of  $\mathcal{K}_{\Gamma,e}^p$  in Eq. (2). Then

$$F_\ell^\alpha(c, \mathbf{h}) = U_\ell^\alpha(\mathbf{h}) + \sum_e \sum_{\substack{r \geq 2, s \geq 1 \\ r+s \in 2\mathbb{Z} \\ rs \leq 2\ell_e}} \frac{\xi_e^{\pi_{r,s}} \kappa_{\Gamma,e}^{(\pi_{r,s})}(\ell) \widehat{\mathcal{R}}_{e;r,s}(\mathbf{h}; \ell)}{c - c_{r,s}(h_e)} F_{\ell - (rs/2)\mathbf{e}_e}^{\top_{r,s} \alpha} \left( c_{r,s}(h_e), \mathbf{h} + \frac{rs}{2} \mathbf{e}_e \right). \quad (50)$$

Here  $\widehat{\mathcal{R}}_{e;r,s}(\mathbf{h}; \ell)$  means the diagonal eigenvalue on the component parity selected by the coefficient  $\ell$ ; it is not a weight-only scalar on a general graph. The shifted coefficient has strictly smaller level on edge  $e$ . Hence the recursion terminates at any fixed multi-level. Half-integral levels retain the lift sign belonging to the removed null level.

### 2.5.3 Generic and coincident poles

Equations (2) and (50) are literal sums of simple poles at generic weights. Here generic means that all Kac zeros encountered along every shifted recursion path are simple and remain distinct from the pole at which the shifted subblock is evaluated.

Coincident pole positions can occur at special weights. They need not be a removable collision of simple terms. The exact theta-graph PBW coefficient at twice-level  $(3, 3, 0)$ , denoted  $D_{330}$  in Section 6, gives a minimal counterexample. After setting  $h_1 = h_0$  and cancelling all common factors, one finds

$$D_{330}|_{h_1=h_0} = \frac{N_{330}(c, h_0, h_\infty)}{4h_0^2((2h_0 + 1)c + 6h_0^2 - 9h_0)^2}, \quad (51)$$

where the numerator is not divisible by the displayed linear factor for generic  $h_0, h_\infty$ . The coincident singularity is therefore a genuine double pole in  $c$ . Merely merging two entries with the same pole key loses this multiplicity and is not a valid exceptional-locus extension.

The present recursion theorem is consequently restricted to the generic locus. Two viable extensions are:

1. analytically detune the weights, assemble the complete generic rational coefficient, and only then take the combined weight limit; or
2. promote the recursion to higher-order Laurent jets, retaining  $(c - c_*)^{-k}$  terms and the derivative data generated by nested shifted subblocks.

Neither extension is implemented here. A finite part should be taken only when the observable or chosen specialization is independently known to be finite; Eq. (51) shows that finiteness is not automatic.

### 3 The regular block at $c = \infty$

#### 3.1 The CCY power-counting template

The global NS algebra is generated by

$$L_{0,\pm 1}, \quad G_{\pm 1/2}, \quad (52)$$

and is  $\mathfrak{osp}(1|2)$ . The non-global creation modes are

$$L_{-n}, \quad n \geq 2, \quad G_{-r}, \quad r \geq \frac{3}{2}. \quad (53)$$

For a superpartition  $A$ , let  $\nu(A)$  count the number of non-global generators in its descendant word. As in CCY, order a basis first by  $\nu$ , then by the mode word. At every fixed level, Gram–Schmidt gives states of the form

$$\hat{\mathcal{L}}_{-A}\nu_h = \hat{\mathcal{O}}_A^{\text{ng}} L_{-1}^n G_{-1/2}^\epsilon \nu_h, \quad n \in \mathbb{Z}_{\geq 0}, \quad \epsilon \in \{0, 1\}, \quad (54)$$

where  $\hat{\mathcal{O}}_A^{\text{ng}}$  contains only the modes in Eq. (53). Its leading component is independent of  $h$ , and it is orthogonal at leading order in  $c$  to a different non-global word. Each contraction of a matched non-global mode contributes one central term, whereas the global algebra has no central extension. Consequently, before converting the tensors to scalar components, one has the ordered super-tensor identities

$$B_h = B_{\text{vac}} \hat{\otimes} B_{h,\mathfrak{osp}(1|2)} + O(c^{-\delta}), \quad (55)$$

$$\rho^\alpha = \rho_{\text{vac}} \hat{\otimes} \rho_{\mathfrak{osp}(1|2)}^\alpha + O(c^{-\delta}). \quad (56)$$

The hat is essential: it is the tensor product in super-vector spaces, so every interchange of a vacuum and a global odd label uses the Koszul braiding. Omitting the hat and reading these as equalities of already ordered scalar arrays was the gap in the earlier derivation. The error is suppressed by some  $\delta > 0$  at the fixed level in question. Here the non-global vacuum factors include their powers of  $c$ ; those powers cancel between the two trinion tensors and the inverse metrics on every sewn edge. A leading vacuum tensor that has no admissible pairing of its bosonic and fermionic non-global modes simply vanishes. The fermions must be put in the canonical half-edge order before this counting is done.

To see why the dependence on the light weights separates, commute a positive non-global mode through a trinion. Its central anticommutator or commutator is the leading term. Terms proportional to  $L_0$ , to another generator, or to a finite weight have one fewer central contraction and are subleading. Thus the leading non-global tensor is precisely the same tensor evaluated in the irreducible NS vacuum module. The leftover  $L_{-1}$  and  $G_{-1/2}$  chains retain all dependence on the weights and on the choice of NS three-point structure. This is the direct super-version of the passage from CCY Eqs. (4.9)–(4.16).

We now perform this reordering explicitly. Concatenate the three ordered slots of every trinion to obtain a half-edge list  $H_V = (h_1, \dots, h_M)$ . Put the same half-edges in contraction order—the

two ends of every internal edge adjacent, followed by the prescribed external order—and let  $\pi_\Gamma$  be the resulting permutation.

The linear part is also read directly from the plumbing maps. Give every half-edge its canonical odd trinion coordinate  $\theta_h^{\text{can}}$ , and record the literal local frame by

$$\theta_h = (-1)^{f_h} \theta_h^{\text{can}}, \quad f_h \in \mathbb{F}_2. \quad (57)$$

For an oriented internal edge  $e = (h, \bar{h})$ , factor out the displayed spin lift  $\xi_e$  and define the residual transition bit  $t_e$  by

$$\theta_{\bar{h}}^{\text{can}} = (-1)^{t_e} i \xi_e \frac{\sqrt{q_e}}{z_h} \theta_h^{\text{can}}. \quad (58)$$

An odd state propagated along this edge therefore sees the computable bit

$$\beta_e = f_h + f_{\bar{h}} + t_e \pmod{2}, \quad \beta_i = f_i \quad \text{for an external half-edge } i. \quad (59)$$

Changing the sign of an intermediate canonical odd coordinate shifts its adjacent  $f$  and  $t$  entries together and leaves Eq. (59) invariant. Thus the inputs are the actual lifted local coordinates, not an additional freely chosen sign table.

For a half-edge parity vector  $p \in \mathbb{F}_2^M$ , compatible with  $p_h = p_{\bar{h}} =: \epsilon_e$  on every internal edge, define

$$Q_\Gamma(p) := \sum_{\substack{i < j \\ \pi_\Gamma(i) > \pi_\Gamma(j)}} p_i p_j + \sum_{e \in E_{\text{int}}} \beta_e \epsilon_e + \sum_{i \in E_{\text{ext}}} \beta_i p_i \pmod{2}. \quad (60)$$

Thus  $\Omega_\Gamma(p) = (-1)^{Q_\Gamma(p)}$ . Equations (57)–(59), followed by the inversion count in Eq. (60), are the promised plumbing-map-to-sign algorithm. They also state exactly where each BPZ or square-root-lift sign is to be entered.

Let  $a(\mathbf{v})$  be the half-edge parity vector of a vacuum level assignment  $\mathbf{v}$ : its two entries on edge  $e$  are  $v_e \pmod{2}$ , and its external entries vanish. Let  $b(\mathbf{r}, \boldsymbol{\eta})$  be the corresponding global vector, with internal entries  $r_e \pmod{2}$  and prescribed external component parities  $\boldsymbol{\eta}$ . The full state has parity  $a + b$ . Polarizing Eq. (60) gives the explicit bilinear form

$$B_\Gamma(a, b) := Q_\Gamma(a + b) + Q_\Gamma(a) + Q_\Gamma(b) \pmod{2} = \sum_{\substack{i < j \\ \pi_\Gamma(i) > \pi_\Gamma(j)}} (a_i b_j + b_i a_j) \pmod{2}. \quad (61)$$

The linear BPZ terms cancel. Functoriality of the Koszul braiding in Eqs. (55)–(56) says precisely that one vacuum/global splitting carries  $(-1)^{B_\Gamma(a, b)}$ .

Expand the separately oriented scalar graph series as

$$\mathcal{V}_{\Gamma, \text{NS}}^{(\infty)} = \sum_{\mathbf{v}} V_{\Gamma, \mathbf{v}}^{(\infty)} \mathbf{x}^{\mathbf{v}}, \quad \mathcal{G}_\Gamma^{\text{osp}(1|2), \alpha} = \sum_{\mathbf{r}} G_{\Gamma, \mathbf{r}}^\alpha \mathbf{x}^{\mathbf{r}}, \quad x_e = \xi_e q_e^{1/2}. \quad (62)$$

Then, for the orientation computed from the lifted plumbing maps, the large- $c$  block has the coefficient formula

$$\boxed{U_{\Gamma, \ell}^{\alpha, \boldsymbol{\eta}} = \sum_{\mathbf{v} + \mathbf{r} = \ell} (-1)^{B_\Gamma(a(\mathbf{v}), b(\mathbf{r}, \boldsymbol{\eta}))} V_{\Gamma, \mathbf{v}}^{(\infty)} G_{\Gamma, \mathbf{r}}^\alpha, \quad \mathcal{U}_\Gamma^{\alpha, \boldsymbol{\eta}} = \sum_{\ell} U_{\Gamma, \ell}^{\alpha, \boldsymbol{\eta}} \mathbf{x}^\ell.} \quad (63)$$

Equation (63) contains no additional undefined product. Its convention-dependent inputs are only the ordered lifted local plumbing maps described above; the endpoint and BPZ/spin-frame signs are their explicit outputs. The coefficient formula is useful for a level-truncated check, but the

regular block itself is the unrestricted all-level sum. Removing the fixed  $\ell$  by  $\sum_{\ell} \sum_{v+r=\ell} = \sum_{v,r}$  gives

$$\boxed{\mathcal{U}_{\Gamma}^{\alpha,\eta}(\mathbf{x}) = \sum_{v,r \in \mathbb{Z}_{\geq 0}^E} (-1)^{B_{\Gamma}(a(v), b(r, \eta))} V_{\Gamma,v}^{(\infty)} G_{\Gamma,r}^{\alpha,\eta} \mathbf{x}^{v+r}.} \quad (64)$$

Here and below  $E = |E_{\text{int}}(\Gamma)|$ . Thus Eq. (64), rather than one fixed  $U_{\Gamma,\ell}$ , is the regular block entering the recursion.

There is also a genuinely function-level version which does not expand the vacuum product coefficient by coefficient. For  $\sigma \in \mathbb{Z}_2^E$ , define the parity projector

$$(P_{\sigma}F)(\mathbf{x}) := 2^{-E} \sum_{\tau \in \mathbb{Z}_2^E} (-1)^{\sigma \cdot \tau} F((( -1)^{\tau_e} x_e)_e). \quad (65)$$

It keeps precisely the monomials whose edge-level parity is  $\sigma$ . Since  $B_{\Gamma}$  is bilinear, set

$$\lambda_{\Gamma,e}(\sigma, \eta) := B_{\Gamma}(a(\mathbf{e}_e), b(\sigma, \eta)) \pmod{2}, \quad (66)$$

where  $\mathbf{e}_e$  is the unit edge-parity vector. Summing the vacuum parity sectors in Eq. (64) then gives the finite parity-resummed formula

$$\boxed{\mathcal{U}_{\Gamma}^{\alpha,\eta}(\mathbf{x}) = \sum_{\sigma \in \mathbb{Z}_2^E} \mathcal{V}_{\Gamma,\text{NS}}^{(\infty)}((( -1)^{\lambda_{\Gamma,e}(\sigma, \eta)} x_e)_e) (P_{\sigma} \mathcal{G}_{\Gamma}^{\text{osp}(1|2), \alpha, \eta})(\mathbf{x}).} \quad (67)$$

This is a sum of  $2^E$  ordinary products of known functions. The first factor is the primitive Schottky product (107), evaluated with the indicated edge-lift sign flips, and the second is the explicit global graph sum (91). Equation (67) is therefore the closed summed regular block; the parity projectors merely implement the displayed finite Fourier sum and introduce no new product operation.

Equivalently, remove the individual orientation signs by setting  $\hat{V}_v = \Omega_{\Gamma}(a(v))V_v$  and  $\hat{G}_r = \Omega_{\Gamma}(b(r, \eta))G_r$ . Then

$$U_{\Gamma,\ell} = \Omega_{\Gamma}(a(\ell) + b(\mathbf{0}, \eta)) \sum_{v+r=\ell} \hat{V}_v \hat{G}_r, \quad (68)$$

which is ordinary coefficient convolution followed by the single full-block orientation sign.

For the theta graph,

$$B_{\Theta}(\bar{\mathbf{v}}, \bar{\mathbf{r}}) = \bar{v}_0 \bar{r}_1 + \bar{r}_0 \bar{v}_1 + \bar{v}_0 \bar{r}_{\infty} + \bar{r}_0 \bar{v}_{\infty} + \bar{v}_1 \bar{r}_{\infty} + \bar{r}_1 \bar{v}_{\infty} \pmod{2}, \quad (69)$$

where bars denote reduction modulo two. This is the complete theta-graph cross sign. In particular,

$$\lambda_{\Theta}(\sigma) = (\sigma_1 + \sigma_{\infty}, \sigma_0 + \sigma_{\infty}, \sigma_0 + \sigma_1) \pmod{2}, \quad (70)$$

so the full theta regular block is the eight-term function-level sum

$$\mathcal{U}_{\Theta}(\mathbf{x}) = \sum_{\sigma \in \mathbb{Z}_2^3} \mathcal{V}_{\Theta}^{(\infty)} \left( (-1)^{\sigma_1 + \sigma_{\infty}} x_0, (-1)^{\sigma_0 + \sigma_{\infty}} x_1, (-1)^{\sigma_0 + \sigma_1} x_{\infty} \right) (P_{\sigma} \mathcal{G}_{\Theta}^{\text{osp}(1|2)})(\mathbf{x}). \quad (71)$$

### 3.2 First non-global check at level 3/2

The first level where the separation is nontrivial has the basis

$$\left\{ G_{-3/2} \nu_h, L_{-1} G_{-1/2} \nu_h \right\}. \quad (72)$$

The exact Gram matrix is

$$B_h^{(3/2)} = \begin{pmatrix} 2h + \frac{2c}{3} & 4h \\ 4h & 2h(2h+1) \end{pmatrix}. \quad (73)$$

For endpoint Ward vectors whose entries remain  $O(1)$ , the contribution of  $G_{-3/2}\nu_h$  is  $O(c^{-1})$ , while the lower-right inverse entry tends to  $[2h(2h+1)]^{-1}$ . Thus only the global state survives. The companion Python script checks this numerically with independent generic Ward vectors.

### 3.3 The genus- $g$ , $N$ -point global $\mathfrak{osp}(1|2)$ network

#### 3.3.1 Global module and edge metric

Use the global basis

$$|h; n, \epsilon\rangle = L_{-1}^n G_{-1/2}^\epsilon \nu_h, \quad \ell = n + \frac{\epsilon}{2}. \quad (74)$$

Since  $G_{-1/2}^2 = L_{-1}$ , no higher fermionic occupation is needed. The raising operators act by

$$\begin{aligned} L_1 |h; n, 0\rangle &= n(2h + n - 1) |h; n - 1, 0\rangle, & G_{1/2} |h; n, 0\rangle &= n |h; n - 1, 1\rangle, \\ L_1 |h; n, 1\rangle &= n(2h + n) |h; n - 1, 1\rangle, & G_{1/2} |h; n, 1\rangle &= (2h + n) |h; n, 0\rangle. \end{aligned} \quad (75)$$

The odd state  $G_{-1/2}\nu_h$  is an  $sl(2)$  primary of weight  $h + 1/2$  and norm  $2h$ . Equation (75) therefore gives the diagonal edge metric

$$\mathcal{N}_h(n, \epsilon) := \langle h; n, \epsilon | h; n, \epsilon \rangle = n! (2h)_{n+\epsilon}. \quad (76)$$

and states of opposite parity are orthogonal.

#### 3.3.2 Plane three-forms and the parity projector

Put slots 1, 2, 3 at  $\infty, 1, 0$ , with weights  $(d_1, d_2, d_3)$ . For  $a, b, c \in \{0, 1\}$ , define

$$T_{k,\ell,m}^{abc}(d_1, d_2, d_3) := \rho \left( G_{-1/2}^a L_{-1}^k \nu_{d_1}, G_{-1/2}^b L_{-1}^\ell \nu_{d_2}, G_{-1/2}^c L_{-1}^m \nu_{d_3} \right). \quad (77)$$

There are two parity-homogeneous NS three-forms. Write their label as  $\kappa \in \{0, 1\}$ , with the same meaning as the earlier vertex label  $\alpha_v$ . The tensor belonging to one fixed structure is

$$T_{k,\ell,m}^{[\kappa];abc} = \delta_{a+b+c \equiv \kappa \pmod{2}} T_{k,\ell,m}^{abc}. \quad (78)$$

This projector is essential: a sphere four-point block in a fixed channel has a separate even or odd internal family, rather than an automatic sum of the two. Belavin–Geiko use component-labelled chiral vertex maps. Before their table is combined with Eq. (78), it must be converted to a fixed-parity trilinear form. The parity of the middle vertex map is  $\kappa + b = a + c \pmod{2}$ , so commuting a supercharge through it carries a sector-dependent sign. Omitting this conversion gives a tensor that fails odd-null factorization.

Normalize the two independent primary structures by  $T_{0,0,0}^{000} = T_{0,0,0}^{010} = 1$ . The global fermionic Ward identities then give the complete zero-bosonic-level table

$$\begin{array}{c|cccccccc} abc & 000 & 100 & 010 & 001 & 110 & 101 & 011 & 111 \\ \hline T_{0,0,0}^{abc} & 1 & 1 & 1 & 1 & B & C & -A & S - \frac{1}{2} \end{array}, \quad (79)$$

where

$$A = d_2 + d_3 - d_1, \quad B = d_1 + d_2 - d_3, \quad C = d_1 - d_2 + d_3, \quad S = d_1 + d_2 + d_3. \quad (80)$$

### 3.3.3 Closed component solution of the Ward identities

The global Ward identities can be solved directly in components, following the three- $L_{-1}$ -chain computation of CCY and retaining a fermionic label on every leg. Distinguish the rising and falling Pochhammer symbols by

$$(x)_n = x(x+1)\cdots(x+n-1), \quad x^n = x(x-1)\cdots(x-n+1). \quad (81)$$

First remove the middle  $L_{-1}$  chain and set

$$t_{k,m}^{abc} := T_{k,0,m}^{abc}. \quad (82)$$

When  $p$  copies of the BPZ-conjugate  $L_1$  act on the third leg,

$$L_1^p L_{-1}^m G_{-1/2}^c \nu_{d_3} = m^p (2d_3 + m - 1 + c)^p L_{-1}^{m-p} G_{-1/2}^c \nu_{d_3}. \quad (83)$$

The binomial choice of which  $L_1$ 's cross the field at one is encoded by

$$\begin{aligned} C_p^{(\delta)}(k, m) &:= \binom{k}{p} m^p (2d_3 + m - 1 + \delta)^p, \\ \mathcal{S}_\delta[f_p] &:= \sum_{p=0}^{\min(k,m)} C_p^{(\delta)}(k, m) f_p. \end{aligned} \quad (84)$$

Commuting the remaining generators through the superprimary at one gives six independent kernels:

$$\begin{aligned} t_{k,m}^{000} &= \mathcal{S}_0[(A)_{m-p}(B+p-m)_{k-p}], \\ t_{k,m}^{010} &= \mathcal{S}_0\left[\left(A + \frac{1}{2}\right)_{m-p}\left(B + \frac{1}{2} + p - m\right)_{k-p}\right], \\ t_{k,m}^{001} &= \mathcal{S}_1\left[\left(A + \frac{1}{2}\right)_{m-p}\left(B - \frac{1}{2} + p - m\right)_{k-p}\right], \\ t_{k,m}^{101} &= C \mathcal{S}_1[(A)_{m-p}(B+p-m)_{k-p}], \\ t_{k,m}^{011} &= -\mathcal{S}_1[(A)_{m-p+1}(B+p-m)_{k-p}], \\ t_{k,m}^{111} &= \left(S - \frac{1}{2}\right) \mathcal{S}_1\left[\left(A + \frac{1}{2}\right)_{m-p}\left(B + \frac{1}{2} + p - m\right)_{k-p}\right]. \end{aligned} \quad (85)$$

The two omitted kernels,  $t^{100}$  and  $t^{110}$ , are fixed without another sum by endpoint reflection:

$$\boxed{t_{k,m}^{abc}(d_1, d_2, d_3) = (-1)^{b(a+c)} t_{m,k}^{cba}(d_3, d_2, d_1), \quad (a, c) = (1, 0).} \quad (86)$$

These formulas are the fixed-parity conversion of the component expressions quoted in Appendix A of Ref. [2]. The binomial choices come from commuting the infinity chain across the insertion, while contractions with the zero chain give the falling factorials in Eq. (83).

It remains to restore  $\ell$  translations of the middle operator. Before those derivatives are applied, scale covariance fixes the coordinate dependence to

$$\begin{aligned} \rho_z(|d_1; k, a\rangle, |d_2; 0, b\rangle, |d_3; m, c\rangle) &= z^{\Lambda_{abc}(k,m)} t_{k,m}^{abc}, \\ \Lambda_{abc}(k, m) &= d_1 - d_2 - d_3 + \frac{a - b - c}{2} + k - m. \end{aligned} \quad (87)$$



Because  $L_{-1}^\ell V(z) = \partial_z^\ell V(z)$ , differentiating Eq. (87) and setting  $z = 1$  gives

$$T_{k,\ell,m}^{abc}(d_1, d_2, d_3) = \left( d_1 - d_2 - d_3 + \frac{a - b - c}{2} + k - m \right)^\ell t_{k,m}^{abc}(d_1, d_2, d_3). \quad (88)$$

This also reduces, at  $a = b = c = 0$ , to CCY Eq. (4.20), since

$$\Lambda_{000}(k, m)^\ell = (d_1 + k - d_2 - \ell + 1 - d_3 - m)_\ell. \quad (89)$$

The argument printed in Eq. (A.6) of Ref. [2] contains an additional  $-\ell$ . That shift cannot obey the elementary check  $\rho(\nu_1, L_{-1}\nu_2, \nu_3) = \partial_z z^{d_1-d_2-d_3}|_{z=1} = d_1 - d_2 - d_3$ . Equation (88) is the translation-covariant version and is the one implemented in the companion code. Equations (78), (85), and (88) give a closed component formula for every global trinion.

### 3.3.4 Graph formula

Fix a canonical ordering of the three half-edges at every vertex. On an internal half-edge use  $a_{v,e} = \epsilon_e$ ; on an external half-edge use the prescribed bottom/top label  $a_{v,i} \in \{0, 1\}$ . Define

$$\Delta_\alpha(\epsilon) := \prod_{v \in V(\Gamma)} \delta_{\sum_{e \ni v} a_{v,e} \equiv \alpha_v \pmod{2}}. \quad (90)$$

Let  $\Omega_\Gamma(\epsilon) = (-1)^{Q_\Gamma(b(\epsilon, \eta))}$ , with  $Q_\Gamma$  given by the explicit half-edge inversion count (60). It is fixed once by the ordered plumbing graph and is not an additional dynamical factor. The global block is

$$\begin{aligned} \mathcal{G}_\Gamma^{\text{osp}(1|2), \alpha}(\mathbf{h}; \mathbf{q}, \boldsymbol{\xi}) = & \sum_{\substack{n_e \geq 0 \\ \epsilon_e \in \{0,1\}}} \left[ \prod_{e \in E_{\text{int}}} \frac{q_e^{n_e + \epsilon_e/2} \xi_e^{\epsilon_e}}{n_e! (2h_e)_{n_e + \epsilon_e}} \right] \\ & \times \Delta_\alpha(\epsilon) \Omega_\Gamma(\epsilon) \prod_{v \in V(\Gamma)} T_{n_{v,1}, n_{v,2}, n_{v,3}}^{a_{v,1} a_{v,2} a_{v,3}}(d_{v,1}, d_{v,2}, d_{v,3}). \end{aligned} \quad (91)$$

For a primary external half-edge,  $n_{v,i} = 0$ . Formula (88) is used after rotating each trinion to the standard  $(\infty, 1, 0)$  slot order. Equation (91) is therefore fully explicit and sparse: each internal edge has only the two states  $(n_e, 0)$  and  $(n_e, 1)$ . The component-normalized block of Ref. [2] is the same expression with  $\Delta_\alpha$  removed.

### 3.3.5 Zero-point blocks and the vacuum channel

For a stable unpunctured surface of genus  $g \geq 2$ , a trivalent pants graph has  $3g - 3$  internal edges and  $2g - 2$  vertices. Specializing Eq. (91) to the absence of external half-edges gives

$$\begin{aligned} \mathcal{U}_{g,0;\Gamma}^\alpha(\mathbf{h}; \mathbf{x}) = & \sum_{\substack{v \in \mathbb{Z}_{\geq 0}^{3g-3}}} V_{\Gamma,v}^{(\infty)} \sum_{\substack{n_e \geq 0 \\ \epsilon_e \in \{0,1\}}} (-1)^{B_\Gamma(a(v), b(2n+\epsilon, 0))} \left[ \prod_{e=1}^{3g-3} \frac{x_e^{v_e + 2n_e + \epsilon_e}}{n_e! (2h_e)_{n_e + \epsilon_e}} \right] \\ & \times \Delta_\alpha(\epsilon) \Omega_\Gamma(\epsilon) \prod_{v=1}^{2g-2} T_{n_{e_{v,1}}, n_{e_{v,2}}, n_{e_{v,3}}}^{\epsilon_{e_{v,1}} \epsilon_{e_{v,2}} \epsilon_{e_{v,3}}}(h_{e_{v,1}}, h_{e_{v,2}}, h_{e_{v,3}}). \end{aligned} \quad (92)$$

Here  $x_e = \xi_e q_e^{1/2}$ , and  $V_{\Gamma, v}^{(\infty)}$  is obtained by expanding the explicit primitive product (107). Thus, after the orientation is compiled from Eqs. (37) and (59), every dynamical symbol on the right-hand side is given either by a finite trinion kernel, one displayed half-edge inversion count, or a coefficient of a displayed infinite product. The slot order and, where needed, the BPZ-rotated version of the second endpoint tensor are fixed by the same lifted plumbing maps as in Eq. (91). Together with Eqs. (85) and (88), the  $\mathfrak{osp}(1|2)$  sum in Eq. (92) is an explicit coefficient formula in that fixed convention. Equation (92) is exactly the  $N = 0$  specialization of Eq. (63); it is not a new prescription.

The vacuum channel is much simpler. Set every internal representation to the irreducible NS vacuum module. One must not insert  $h_e = 0$  directly into the generic denominator in Eq. (92); instead impose the quotient (96). It removes every global descendant, so that

$$\mathcal{G}_{g,0}^{\mathfrak{osp}(1|2),\text{vac}} = 1, \quad \mathcal{U}_{g,0}^{\text{vac},\delta} = \mathcal{V}_{g,\text{NS}}^{(\infty),\delta} = \prod_{[\gamma] \in \mathcal{P}_{\text{un}}(\mathfrak{S}_{\Gamma})} \prod_{m=2}^{\infty} \frac{1 + u_{\gamma} k_{\gamma}^{m-1}}{1 - k_{\gamma}^m}, \quad u_{\gamma}^2 = k_{\gamma}. \quad (93)$$

Equation (93), together with the definitions and derivation in Section 3.4, is the promised explicit  $N = 0$  specialization. It is also the sharp consistency test that prevents a merely formal sewing prescription from being mistaken for a closed regular block.

The two unstable low-genus cases provide useful normalizations. With the primary and Casimir powers stripped, the sphere zero-point block is one. For a torus and a generic internal NS Verma module,

$$\mathcal{U}_{1,0}(h; q, \xi) = \frac{1 + \xi q^{1/2}}{1 - q} \mathcal{V}_{1,\text{NS}}^{(\infty)}(q; \xi) = \prod_{n=1}^{\infty} \frac{1 + \xi q^{n-1/2}}{1 - q^n}. \quad (94)$$

For the irreducible vacuum module, the global prefactor in the middle expression is removed and one recovers Eq. (109).

### 3.4 The universal NS vacuum network

#### 3.4.1 Sewing definition and the free large- $c$ vacuum module

The vacuum coefficients in Eq. (63) are not optional at positive genus. Define their generating function in the same plumbing frame by

$$\mathcal{V}_{\Gamma,\text{NS}}^{(\infty)}(\mathbf{q}; \boldsymbol{\xi}) := \lim_{c \rightarrow \infty} \mathcal{F}_{\Gamma}^{\text{vac}}(c; \mathbf{q}, \boldsymbol{\xi}), \quad (95)$$

where every internal and external module is the irreducible NS vacuum module,

$$L_{-1}|0\rangle = 0, \quad G_{-1/2}|0\rangle = 0. \quad (96)$$

Equation (95), evaluated coefficient by coefficient using Eq. (17), is a tautologically exact sewing definition. We now evaluate it by the same link strategy that gives the bosonic CCY answer.

Normalize the non-global vacuum oscillators by

$$a_{-n} := \sqrt{\frac{12}{c n(n^2 - 1)}} L_{-n}, \quad n \geq 2, \quad \psi_{-r} := \sqrt{\frac{3}{c(r^2 - \frac{1}{4})}} G_{-r}, \quad r \geq \frac{3}{2}. \quad (97)$$

The algebra (5) gives, inside every fixed-particle-number matrix element,

$$[a_n, a_{-m}] = \delta_{nm} + O(c^{-1/2}), \quad \{\psi_r, \psi_{-s}\} = \delta_{rs} + O(c^{-1/2}), \quad [a_n, \psi_{-r}] = O(c^{-1/2}). \quad (98)$$

The error terms contain a normalized noncentral generator. They vanish as  $c \rightarrow \infty$ , while the central contractions remain finite. The two one-particle global  $sl(2)$  modules are

$$\mathcal{H}_T = \text{span}\{L_{-1}^j L_{-2}|0\rangle : j \geq 0\}, \quad \mathcal{H}_G = \text{span}\{L_{-1}^j G_{-3/2}|0\rangle : j \geq 0\}. \quad (99)$$

Consequently the normalized large- $c$  vacuum module is, coefficient by coefficient,

$$\mathcal{H}_{\text{vac}}^{(\infty)} = \text{Sym } \mathcal{H}_T \otimes \Lambda \mathcal{H}_G. \quad (100)$$

This is the only new representation-theoretic input relative to CCY: the stress-tensor particles are symmetric, whereas the supercurrent particles form an exterior Fock space.

### 3.4.2 Lifted Schottky words

Choose the Schottky marking associated to the CCY plumbing frame and write its rank- $g$  group as  $\mathfrak{S}_\Gamma \subset PSL(2, \mathbb{C})$ . The NS spin data select one of the two  $SL(2, \mathbb{C})$  lifts of every generator. Denote the resulting lifted group by  $\tilde{\mathfrak{S}}_{\Gamma, \delta}$ , where  $\delta$  is the spin-structure label. These are the spin structures compatible with the chosen all-NS handlebody marking; a sector with a Ramond cycle requires a different state space and is outside the present formula. For a loxodromic word  $\gamma$ , let a lifted representative have eigenvalues  $\lambda_\gamma, \lambda_\gamma^{-1}$ , with  $|\lambda_\gamma| > 1$ , and define

$$u_\gamma := \lambda_\gamma^{-1}, \quad k_\gamma := u_\gamma^2, \quad |k_\gamma| < 1. \quad (101)$$

Here  $k_\gamma$  is the ordinary attracting Schottky multiplier and  $u_\gamma$  is its spin-refined half-multiplier. Replacing the lifted word by its negative flips  $u_\gamma$  but leaves  $k_\gamma$  fixed. Cyclic conjugation and inversion leave both attracting multipliers unchanged when the inverse word is lifted as the matrix inverse. Thus  $u_\gamma$ , not an independently chosen symbol  $\pm\sqrt{k_\gamma}$ , is the unambiguous datum needed by the fermionic block.

The CCY sewing correlator may be reorganized into closed contraction links. Oriented primitive links are in one-to-one correspondence with primitive Schottky conjugacy classes, and an  $s$ -fold traversal gives the  $s$ -th power of the corresponding word [5]. The one-particle traces for such a traversal are

$$Z_T(\gamma^s) = \sum_{j=0}^{\infty} k_\gamma^{s(j+2)} = \frac{k_\gamma^{2s}}{1 - k_\gamma^s}, \quad (102)$$

$$Z_G(\tilde{\gamma}^s) = \sum_{j=0}^{\infty} u_\gamma^{s(2j+3)} = \frac{u_\gamma^{3s}}{1 - k_\gamma^s}. \quad (103)$$

The exponents  $2 + j$  and  $3/2 + j$  are the weights of  $\partial^j T$  and  $\partial^j G$ . The latter is written with the  $SL(2)$  lift, so a weight- $3/2 + j$  state contributes  $u_\gamma^{3+2j}$ .

### 3.4.3 Link resummation in the Schottky frame

For one unoriented primitive class, the usual cyclic symmetry of an  $s$ -fold link supplies  $1/s$ . An  $s$ -cycle of odd particles has fermion parity  $(-1)^{s-1}$ . Hence the connected link sum is

$$\log \mathcal{V}_\gamma = \sum_{s=1}^{\infty} \frac{1}{s} \left[ \frac{k_\gamma^{2s}}{1 - k_\gamma^s} + (-1)^{s+1} \frac{u_\gamma^{3s}}{1 - k_\gamma^s} \right]. \quad (104)$$

The relative sign is equivalently the finite-dimensional identity

$$\det(1 + K) = \exp \left[ \sum_{s \geq 1} \frac{(-1)^{s+1}}{s} \operatorname{tr} K^s \right], \quad (105)$$

applied to the supercurrent one-particle kernel. It is therefore fixed by the exterior Fock space, rather than imported from a putative supergravity answer. Expanding the denominators in Eq. (104) and summing over  $s$  gives

$$\log \mathcal{V}_\gamma = - \sum_{m=2}^{\infty} \log(1 - k_\gamma^m) + \sum_{m=2}^{\infty} \log(1 + u_\gamma k_\gamma^{m-1}). \quad (106)$$

Let  $\mathcal{P}_{\text{un}}(\mathfrak{S}_\Gamma)$  denote primitive conjugacy classes with  $\gamma$  and  $\gamma^{-1}$  identified. The resulting Schottky-frame one-loop determinant is

$$\mathcal{V}_{\Gamma, \text{NS}}^{(\infty), \delta} = \prod_{[\gamma] \in \mathcal{P}_{\text{un}}(\mathfrak{S}_\Gamma)} \prod_{m=2}^{\infty} \frac{1 + u_\gamma k_\gamma^{m-1}}{1 - k_\gamma^m}, \quad u_\gamma^2 = k_\gamma. \quad (107)$$

Near the plumbing cusp this is a convergent Schottky product. Its leading theta terms and all coefficients through total level eight agree with the independent ordered plumbing calculation. This is the CCY-frame vacuum input itself; there is no additional frame-conversion operator. The earlier apparent mismatch came from using an incorrect second theta Schottky generator, as made explicit below. CCY identify the primitive product with the vacuum block in the plumbing frame [3], and the explicit genus-two generators and their direct descendant check are given in Ref. [6]. In the oriented-class convention used in CCY Eq. (5.5), the same formula is

$$\mathcal{V}_{\Gamma, \text{NS}}^{(\infty), \delta} = \prod_{\gamma \in \mathcal{P}_{\text{or}}} \prod_{m=2}^{\infty} (1 - k_\gamma^m)^{-1/2} (1 + u_\gamma k_\gamma^{m-1})^{1/2}. \quad (108)$$

The two orientations have the same multiplier and combine into Eq. (107). Dropping  $\mathcal{H}_G$  removes the numerator and reproduces the bosonic Schottky determinant used by CCY. The fermionic numerator also agrees with the independent genus-one gravitino determinant [7]; this is a check, not an input to the derivation.

#### 3.4.4 Genus-zero and genus-one normalizations

At genus zero the Schottky group has no nontrivial primitive class, so the empty product gives  $\mathcal{V}_{0, \text{NS}}^{(\infty)} = 1$ . At genus one there is one unoriented primitive generator. Writing  $k_\gamma = q$  and  $u_\gamma = \xi q^{1/2}$  gives, with the Casimir factor removed,

$$\mathcal{V}_{1, \text{NS}}^{(\infty)}(q; \xi) = \prod_{m=2}^{\infty} \frac{1 + \xi q^{m-1/2}}{1 - q^m}. \quad (109)$$

The numerator begins at  $G_{-3/2}$ , and the denominator at  $L_{-2}$ . The choices  $\xi = +1$  and  $-1$  give the ungraded NS and  $\widetilde{\text{NS}}$  traces.

## 4 Superspace repackaging

This section is not needed for the derivation in Section 3. It repackages the already established component answer and makes the  $L_{-1} - G_{-1/2}$  relation and the Koszul signs more economical.

#### 4.1 The superspace trinion

The table above and all its descendants follow from one superspace three-point function. For a superprimary  $\Phi_i(z_i, \theta_i) = \phi_i(z_i) + \theta_i \psi_i(z_i)$ , introduce

$$Z_{ij} := z_i - z_j - \theta_i \theta_j \quad (110)$$

and the odd superprojective invariant

$$\eta_{123} := \frac{Z_{23}\theta_1 + Z_{31}\theta_2 + Z_{12}\theta_3 + \theta_1\theta_2\theta_3}{(Z_{12}Z_{13}Z_{23})^{1/2}}. \quad (111)$$

The two homogeneous three-point structures are the coefficients of (1) and  $\eta_{123}$  in

$$\langle \Phi_1 \Phi_2 \Phi_3 \rangle = \frac{C_{123} + \tilde{C}_{123} \eta_{123}}{Z_{12}^B Z_{13}^C Z_{23}^A}. \quad (112)$$

These conventions agree with the standard  $\mathfrak{osp}(1|2)$  superspace correlator of Ref. [8]. The normalizations of  $C_{123}$  and  $\tilde{C}_{123}$  are kept independent, as required for the even and odd trinion structures.

Use BPZ local supercoordinates about  $(\infty, 1, 0)$ ,

$$z_1 = -u^{-1}, \quad \theta_1 = z_1 \vartheta_1, \quad z_2 = 1 + v, \quad \theta_2 = \vartheta_2, \quad z_3 = w, \quad \theta_3 = \vartheta_3, \quad (113)$$

together with the superprimary Jacobian in the first slot, and define

$$X := 1 + u(1 + v), \quad Y := 1 + uw, \quad Z := 1 + v - w. \quad (114)$$

There is one convention conversion before sewing. If  $N_i = \vartheta_i \partial_{\vartheta_i}$ , the component-labelled vertex map of the superspace correlator is converted to our fixed-parity trilinear form by

$$\mathcal{R}_{\text{tri}} := \exp[i\pi N_2(1 - N_1)], \quad \mathcal{R}_{\text{tri}} : T_{\text{sf}}^{abc} \mapsto (-1)^{b(1-a)} T_{\text{sf}}^{abc}. \quad (115)$$

This is exactly the component-map/trilinear-form distinction used below Eq. (78); it is not an additional physical sign.

After substituting Eq. (113) into Eq. (112), applying Eq. (115), and expanding in the ordered monomials  $\vartheta_1^a \vartheta_2^b \vartheta_3^c$ , all eight component functions collapse to

$$\boxed{\mathcal{T}^{abc}(u, v, w) = \kappa_{abc} X^{-B-(a+b-c)/2} Y^{-C-(a+c-b)/2} Z^{-A-(b+c-a)/2}}. \quad (116)$$

The only remaining data are the primary coefficients

$$\begin{array}{c|cccccccc} abc & 000 & 100 & 010 & 001 & 110 & 101 & 011 & 111 \\ \hline \kappa_{abc} & 1 & 1 & 1 & 1 & B & C & -A & S - \frac{1}{2}. \end{array} \quad (117)$$

Thus the three half-integer exponent shifts are not case-by-case Ward recursions: they are simply the incidence combinations  $(a + b - c)/2$ ,  $(a + c - b)/2$ , and  $(b + c - a)/2$  dictated by the three superdistances.

For completeness, combine the eight coefficients into the Grassmann polynomial

$$\widehat{\mathcal{T}}(u, v, w; \boldsymbol{\vartheta}) := \sum_{a,b,c=0}^1 \vartheta_1^a \vartheta_2^b \vartheta_3^c \mathcal{T}^{abc}(u, v, w). \quad (118)$$

The local odd lowering operators may be represented by

$$\mathcal{D}_1 := \partial_{\vartheta_1} - \vartheta_1 \partial_u, \quad \mathcal{D}_2 := \partial_{\vartheta_2} + \vartheta_2 \partial_v, \quad \mathcal{D}_3 := \partial_{\vartheta_3} + \vartheta_3 \partial_w, \quad (119)$$

so that  $\mathcal{D}_1^2 = -\partial_u$ ,  $\mathcal{D}_2^2 = \partial_v$ , and  $\mathcal{D}_3^2 = \partial_w$ . The minus sign in the first identity is the BPZ inversion in Eq. (113). With left Grassmann derivatives and the displayed slot order, the entire global trinion is therefore

$$T_{k,\ell,m}^{abc}(d_1, d_2, d_3) = \mathcal{D}_3^{2m+c} \mathcal{D}_2^{2\ell+b} \mathcal{D}_1^{2k+a} \widehat{\mathcal{T}}(u, v, w; \vartheta) \Big|_{u=v=w=\vartheta=0}. \quad (120)$$

Equivalently, extracting a fixed Grassmann coefficient first gives the particularly economical bosonic formula

$$T_{k,\ell,m}^{abc} = (-1)^k \partial_u^k \partial_v^\ell \partial_w^m \mathcal{T}^{abc}(u, v, w) \Big|_{u=v=w=0}. \quad (121)$$

Equations (116) and (121) are the desired closed product form of the  $\mathfrak{osp}(1|2)$  trinion. In particular, they define the supervertex in Eq. (122) without referring to an abstract Ward recursion or an unspecified product.

## 4.2 Graph sewing as a Berezin integral

The discrete sum over the edge labels  $\epsilon_e$  and its orientation sign can be absorbed into a Grassmann integral. Assign a Grassmann variable  $\theta_h$  to every internal half-edge. For a trinion  $v$ , package the component kernels into the supervertex

$$\mathbb{T}_v^{\alpha_v}(\mathbf{n}_v; \boldsymbol{\theta}_v) := \sum_{\substack{a_{v,i} \in \{0,1\}, \ a_{v,i} = \eta_i \text{ on external slots} \\ \sum_i a_{v,i} = \alpha_v \pmod{2}}} T_{n_{v,1}, n_{v,2}, n_{v,3}}^{a_{v,1}, a_{v,2}, a_{v,3}}(d_{v,1}, d_{v,2}, d_{v,3}) \prod_{i: (v,i) \in H_{\text{int}}(\Gamma)} \theta_{v,i}^{a_{v,i}}. \quad (122)$$

On an external half-edge the prescribed component  $\eta_i$  is inserted in  $a_{v,i}$ , and no integration variable is introduced. This is the component expansion of the usual  $\mathfrak{osp}(1|2)$  superspace trinion; the product formula (116) and the superderivative extraction (120) give all its coefficients directly.

Suppose the internal edge  $e$  joins half-edges  $h_e, \bar{h}_e$ . At fixed external parity  $\boldsymbol{\eta}$ , write

$$\beta_{\Gamma,e}(\boldsymbol{\eta}) := \beta_e + B_\Gamma(b(\mathbf{e}_e, \mathbf{0}), b(\mathbf{0}, \boldsymbol{\eta})), \quad \Omega_{\Gamma,\text{ext}}(\boldsymbol{\eta}) := (-1)^{Q_\Gamma(b(\mathbf{0}, \boldsymbol{\eta}))}. \quad (123)$$

Here  $\beta_e$  is the frame-derived BPZ/spin bit of Eq. (59). At fixed bosonic occupation  $n_e$ , define

$$\mathbb{K}_{e,n_e}(\theta_{h_e}, \theta_{\bar{h}_e}) := \theta_{h_e} \theta_{\bar{h}_e} + (-1)^{\beta_{\Gamma,e}(\boldsymbol{\eta})} \frac{x_e}{2h_e + n_e}, \quad x_e = \xi_e q_e^{1/2}. \quad (124)$$

With the convention

$$\int_\Gamma^{\text{ord}} D\theta \theta_{\pi_\Gamma(1)} \cdots \theta_{\pi_\Gamma(2E)} = 1, \quad (125)$$

the measure is the reverse Berezin differential in contraction order. Expanding one edge immediately gives

$$\int d\theta_{\bar{h}_e} d\theta_{h_e} \mathbb{K}_{e,n_e}(A_0 + \theta_{h_e} A_1)(B_0 + \theta_{\bar{h}_e} B_1) = A_0 B_0 + (-1)^{\beta_{\Gamma,e}(\boldsymbol{\eta})} \frac{x_e}{2h_e + n_e} A_1 B_1. \quad (126)$$

Thus the integral forces the endpoint parities to agree. Multiplication by  $q_e^{n_e}/[n_e!(2h_e)_{n_e}]$  gives respectively

$$\frac{q_e^{n_e}}{n_e!(2h_e)_{n_e}}, \quad (-1)^{\beta_{\Gamma,e}(\boldsymbol{\eta})} \frac{\xi_e q_e^{n_e+1/2}}{n_e!(2h_e)_{n_e+1}}, \quad (127)$$

which are exactly the even and odd inverse global Gram factors.

Order the product of supervertices in vertex-slot order and the Berezin measure in contraction order. The sign required to move the surviving odd variables between these two orders is precisely the inversion polynomial  $Q_\Gamma$ . Consequently Eq. (91) is equivalently the single superspace sewing integral

$$\mathcal{G}_\Gamma^{\text{osp}(1|2),\alpha}(\mathbf{h}; \mathbf{q}, \boldsymbol{\xi}) = \Omega_{\Gamma, \text{ext}}(\boldsymbol{\eta}) \sum_{\mathbf{n} \in \mathbb{Z}_{\geq 0}^E} \left[ \prod_{e \in E_{\text{int}}} \frac{q_e^{n_e}}{n_e! (2h_e)_{n_e}} \right] \int_\Gamma^{\text{ord}} D\theta \left[ \prod_{e \in E_{\text{int}}} \mathbb{K}_{e, n_e} \right] \left[ \prod_{v \in V(\Gamma)} \mathbb{T}_v^{\alpha_v} \right]. \quad (128)$$

There is no separate  $\epsilon$ -sum or manually appended  $\Omega_\Gamma$  in Eq. (128); both are performed by the Berezin integral.

The same variables also absorb the vacuum/global cross sign. Choose one endpoint  $h_e$  of each edge and let

$$\hat{N}_e := \theta_{h_e} \frac{\partial}{\partial \theta_{h_e}} \quad (129)$$

act only on the product of supervertices. Edge contraction makes the result independent of the chosen endpoint. Define the binary matrix and external shift

$$M_{\Gamma, ef} := B_\Gamma(a(\mathbf{e}_e), b(\mathbf{e}_f, \mathbf{0})), \quad \mu_{\Gamma, e}(\boldsymbol{\eta}) := B_\Gamma(a(\mathbf{e}_e), b(\mathbf{0}, \boldsymbol{\eta})), \quad (130)$$

and the commuting even operators

$$\hat{\lambda}_{\Gamma, e} := \mu_{\Gamma, e}(\boldsymbol{\eta}) + \sum_f M_{\Gamma, ef} \hat{N}_f \pmod{2}. \quad (131)$$

On a component with edge parity  $\epsilon$ , this has eigenvalue  $\lambda_{\Gamma, e}(\epsilon, \boldsymbol{\eta})$  of Eq. (66). The fully summed regular block therefore has the compact Berezin form

$$\mathcal{U}_\Gamma^{\alpha, \boldsymbol{\eta}}(\mathbf{x}) = \Omega_{\Gamma, \text{ext}}(\boldsymbol{\eta}) \sum_{\mathbf{n} \in \mathbb{Z}_{\geq 0}^E} \left[ \prod_e \frac{q_e^{n_e}}{n_e! (2h_e)_{n_e}} \right] \int_\Gamma^{\text{ord}} D\theta \left[ \prod_e \mathbb{K}_{e, n_e} \right] \times \mathcal{V}_{\Gamma, \text{NS}}^{(\infty)} \left( (e^{i\pi \hat{\lambda}_{\Gamma, e}} x_e)_e \right) \left[ \prod_v \mathbb{T}_v^{\alpha_v} \right]. \quad (132)$$

The number operators in the vacuum factor act only on the supervertex product to their right. Expanding that factor on a parity eigencomponent gives  $(-1)^{B_\Gamma(a(v), b(\epsilon, \boldsymbol{\eta}))}$  for every vacuum monomial  $\mathbf{x}^v$ ; hence Eq. (132) is algebraically identical to Eqs. (64) and (67). It is the superspace version of the graded product: all odd labels are integrated rather than explicitly summed.

## 5 Exploratory super-Riemann-surface plumbing

The following construction is recorded for later use but is not an input to the recursion. It is a local reformulation of the component block, not an independent derivation of the regular term. In particular, an NS tube has an even smoothing parameter; the genuine odd coordinates belong to the three-punctured super-spheres.

## 5.1 Odd moduli in an NS pants decomposition

The state-sewing formula has an intrinsic super-Riemann-surface interpretation. For a stable genus- $g$  surface with  $N$  NS punctures,

$$\dim \mathfrak{M}_{g,N}^{\text{NS}} = (3g - 3 + N \mid 2g - 2 + N). \quad (133)$$

A trivalent pants graph has precisely

$$|E_{\text{int}}| = 3g - 3 + N, \quad |V| = 2g - 2 + N. \quad (134)$$

The matching is geometric: every internal NS tube contributes the even coordinate  $q_e$ , whereas every three-NS super-pants contributes one odd coordinate  $\nu_v$ . The NS node itself has no independent odd smoothing parameter. Sewing two NS punctures with Eq. (11) removes two punctures and introduces one even modulus, leaving the total odd dimension unchanged. These facts follow from the standard deformation theory of NS punctures and NS degenerations [9, 4].

Locally, an NS puncture at the origin of a superdisc  $(w, \vartheta)$  can be moved to  $(t, \nu)$  by

$$z = w + t + \vartheta\nu, \quad \theta = \vartheta + \nu. \quad (135)$$

Consequently a superprimary insertion obeys

$$\Phi(t, \nu) = (1 + \nu G_{-1/2})\Phi(t). \quad (136)$$

Equations (135) and (136) make precise the sense in which the even and odd local motions are treated equally: their generators are  $L_{-1}$  and  $G_{-1/2}$ , with  $G_{-1/2}^2 = L_{-1}$ .

Choose an ordering  $v = 1, \dots, V$  of the trinions and normalize the surface-state generating form by

$$\rho_v(\nu_v) := \rho_v^0 + \nu_v \rho_v^1. \quad (137)$$

For an internal module, define the full sewing propagator

$$P_{h,c}(q, \xi) := \sum_{\ell \in \frac{1}{2}\mathbb{Z}_{\geq 0}} \xi^{2\ell} q^\ell \sum_{|I|=|J|=\ell} |\mathcal{L}_I \nu_h\rangle (B_{h,c}^{(\ell)})^{-1} [I, J] \langle \mathcal{L}_J \nu_h|. \quad (138)$$

Since  $\xi = \pm 1$ , the factor  $\xi^{2\ell}$  is one at integral level and  $\xi$  at half-integral level. The unintegrated supermoduli block is the single Grassmann-valued sewing contraction

$$\mathbb{F}_\Gamma(c, \mathbf{h}; \mathbf{q}, \boldsymbol{\xi}; \boldsymbol{\nu}) := \text{Sew}_\Gamma \left[ \bigotimes_{v=1}^V \rho_v(\nu_v), \bigotimes_{e \in E_{\text{int}}} P_{h_e, c}(q_e, \xi_e) \right]. \quad (139)$$

The tensor products and contractions use the same vertex-slot and edge orders as Eq. (17). Expanding in the genuine odd moduli gives

$$\mathbb{F}_\Gamma(\boldsymbol{\nu}) = \sum_{\boldsymbol{\alpha} \in \{0,1\}^V} \nu_1^{\alpha_1} \dots \nu_V^{\alpha_V} \mathcal{F}_\Gamma^\alpha. \quad (140)$$

This expansion can be inverted by an actual Berezin integral. Set

$$\sigma(\boldsymbol{\alpha}) := \sum_{1 \leq i < j \leq V} \alpha_i (1 - \alpha_j), \quad \nu^{\vee \boldsymbol{\alpha}} := (-1)^{\sigma(\boldsymbol{\alpha})} \prod_{i: \alpha_i=0}^{\nearrow} \nu_i, \quad (141)$$



and use the measure  $D\nu = d\nu_V \cdots d\nu_1$ , normalized by  $\int D\nu \nu_1 \cdots \nu_V = 1$ . Then

$$\boxed{\mathcal{F}_\Gamma^\alpha = \int D\nu \nu^{\vee\alpha} \mathbb{F}_\Gamma(\nu).} \quad (142)$$

The sign in Eq. (141) is the unique one that moves the complementary monomial past  $\nu^\alpha$  into canonical order.

The propagator treats all NS modes uniformly. On a PBW state,

$$q^{L_0-h} \prod_{n \geq 1} L_{-n}^{k_n} \prod_{r \in \mathbb{Z}_{>0}-1/2} G_{-r}^{\epsilon_r} \nu_h = q^{\sum_n n k_n + \sum_r r \epsilon_r} \prod_{n \geq 1} L_{-n}^{k_n} \prod_r G_{-r}^{\epsilon_r} \nu_h. \quad (143)$$

Thus no additional tube variable is needed to weight fermionic descendants. Equivalently, there is no  $G_0$  in an NS module with which one could form a tube insertion  $e^{\eta G_0}$ . An odd coordinate attached to one end of an NS tube is a choice of where to place an adjacent super-pants modulus, not a new modulus for every edge. In contrast, a Ramond degeneration does carry an odd gluing modulus and a  $G_0$  insertion; that distinction is outside the all-NS problem.

Equation (142) is a kinematic coefficient extraction for the chiral matter block. A physical superstring integral contains the superghost integral form  $\delta(B_{\nu_v})$ , picture assignments, and possibly vertical corrections. Therefore an unweighted  $\int D\nu$  should not be identified with a physical amplitude inside the matter block alone.

This construction is local to the chosen plumbing chart. On overlaps of supermoduli charts, changing the representatives of the odd coordinates can induce nilpotent shifts

$$q'_e = q_e + O(\nu_v \nu_w), \quad (144)$$

as required by the  $G(z)G(0) \sim 2T(0)/z$  operator product. Thus Eq. (139) is not a claim that supermoduli space is globally split. It is a local formal coordinate system near the fixed CCY pants degeneration, which is precisely the domain in which the plumbing expansion and its  $c$ -recursion are defined.

## 5.2 Relation to the regular block

The genuine supermoduli can now be included without returning to components. Do not identify them with the half-edge variables  $\theta_h$ . The latter are auxiliary contraction variables, whereas each  $\nu_v$  is a coordinate on the supermoduli space. Define the unprojected super-pants vertex

$$\mathbb{T}_v(\nu_v; \mathbf{n}_v, \boldsymbol{\theta}_v) := \mathbb{T}_v^0(\mathbf{n}_v; \boldsymbol{\theta}_v) + \nu_v \mathbb{T}_v^1(\mathbf{n}_v; \boldsymbol{\theta}_v). \quad (145)$$

Replacing every projected vertex in Eq. (132) gives the regular generating superblock

$$\boxed{\begin{aligned} \mathcal{U}_\Gamma(\nu; \mathbf{x}) = \Omega_{\Gamma, \text{ext}}(\boldsymbol{\eta}) \sum_{\mathbf{n} \in \mathbb{Z}_{\geq 0}^E} \left[ \prod_e \frac{q_e^{n_e}}{n_e! (2h_e)_{n_e}} \right] \int_\Gamma^{\text{ord}} D\theta \left[ \prod_e \mathbb{K}_{e, n_e} \right] \\ \times \mathcal{V}_{\Gamma, \text{NS}}^{(\infty)} \left( (e^{i\pi \widehat{\lambda}_{\Gamma, e}} x_e)_e \right) \left[ \prod_{v=1}^V \mathbb{T}_v(\nu_v; \mathbf{n}_v, \boldsymbol{\theta}_v) \right]. \end{aligned}} \quad (146)$$

It contains the whole  $2^V$ -component family. Performing the true supermoduli integral after the auxiliary edge contractions gives

$$\boxed{\mathcal{U}_\Gamma^\alpha(\mathbf{x}) = \int D\nu \nu^{\vee\alpha} \mathcal{U}_\Gamma(\nu; \mathbf{x}).} \quad (147)$$

Thus the global  $L_{-1}, G_{-1/2}$  descendants, the edge parity sum, the Koszul signs, and the trinion-sector extraction are all performed within one ordered superspace expression. The two Berezin integrations have different geometric status and should remain notationally distinct.

### 5.3 Scope of the plumbing proposal

The recursion computes a holomorphic algebraic NS conformal block. A type 0B string amplitude additionally requires

antiholomorphic block + GSO/spin sum + structure constants + superghosts + supermoduli measure. (148)

In particular, an NS trinion has an odd modulus. Equation (140) identifies  $\alpha_v$  as its Grassmann coefficient, and Eq. (142) extracts it by Berezin integration. A physical picture or odd-modulus prescription supplies the superghost integral form and selects the appropriate contraction. The matter block alone must not replace that prescription by an unweighted sum over  $\alpha_v$ .

The string-note plumbing formula (11) explains why the reduced bosonic number  $q_e$  is insufficient for half-integral levels. A code key must contain the lift  $\xi_e$  or an equivalent spin-structure identifier. Two blocks evaluated at the same complex  $q_e$  can differ by the signs of all odd-level terms.

## 6 Consistency checks

The derivation above is checked first in simple channels and then against a direct finite- $c$  PBW construction. The direct block and the recursive block are generated independently.

### 6.1 Sphere four-point block

There is one internal edge and no positive-genus vacuum factor. The recursion becomes

$$\mathcal{F}_h(c; z) = \mathcal{G}_h^{\text{osp}(1|2)}(z) + \sum_{\substack{r \geq 2, s \geq 1 \\ r+s \in 2\mathbb{Z}}} \frac{z^{rs/2} J_{r,s}(h) A_{r,s}(h) \Sigma_{L;r,s} \Sigma_{R;r,s}}{c - c_{r,s}(h)} \mathcal{F}_{h+rs/2}(c_{r,s}(h); z). \quad (149)$$

For bottom external components the seed is Eq. (205). Top components replace the two endpoint global Ward factors, but do not alter the pole positions or null norm.

#### 6.1.1 Numerical $c$ - versus $h$ -recursion test at $\hat{c} = 9$

As an independent numerical test of Eq. (149), we compared its central-charge recursion with a fixed- $b$ , internal-weight recursion implemented directly from Ref. [1]. The comparison was made separately for the two parity blocks

$$H_e(q) = \sum_{n \geq 0} H_n q^n, \quad H_o(q) = \sum_{n \geq 0} H_{n+\frac{1}{2}} q^{n+\frac{1}{2}}. \quad (150)$$

These are the integer-level and half-integer-level NS families, respectively. They should not be confused with evaluating the spin lift at  $\xi = +1$  or  $-1$ . If the two families are packaged with a common kinematic normalization, then

$$H(q, \xi) = H_e(q) + \xi H_o(q), \quad H_e = \frac{H(q, +1) + H(q, -1)}{2}, \quad H_o = \frac{H(q, +1) - H(q, -1)}{2}. \quad (151)$$

Thus  $\xi = +1$  gives the sum of the two parity sectors and  $\xi = -1$  their signed difference. On the sphere plumbing tree the lift on the single internal tube is a convention rather than an invariant spin holonomy; only its parity projection is being used here.

The benchmark is the self-dual super-Liouville point

$$\hat{c} = 9, \quad c = \frac{27}{2}, \quad b = 1, \quad Q = b + b^{-1} = 2, \quad (152)$$

with the momentum convention

$$h(P) = \frac{1}{2} \left( \frac{Q^2}{4} + P^2 \right). \quad (153)$$

We chose

$$\begin{aligned} (P_1, P_2, P_3, P_4) &= \left( \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{3}{5} \right), & P_{\text{int}} &= \frac{7}{10}, \\ (h_1, h_2, h_3, h_4) &= \left( \frac{5}{8}, \frac{5}{9}, \frac{17}{32}, \frac{17}{25} \right), & h_{\text{int}} &= \frac{149}{200}. \end{aligned} \quad (154)$$

The order-ten cutoff means that the even elliptic block retains  $q^0, \dots, q^{10}$ , whereas the odd block retains  $q^{1/2}, \dots, q^{19/2}$ . The common elliptic prefactor was restored before comparing the full chiral blocks. For each parity define

$$\delta_{e,o}(z) = \frac{|\mathcal{F}_{e,o}^{(c)}(z) - \mathcal{F}_{e,o}^{(h)}(z)|}{\max(|\mathcal{F}_{e,o}^{(c)}(z)|, |\mathcal{F}_{e,o}^{(h)}(z)|)}. \quad (155)$$

The ten-point comparison is

| $z$  | $q(z)$       | $\mathcal{F}_e^{(c)}$ | $\delta_e$             | $\mathcal{F}_o^{(c)}$ | $\delta_o$             |
|------|--------------|-----------------------|------------------------|-----------------------|------------------------|
| 0.05 | 0.0032057870 | 3.738629098464        | $2.63 \times 10^{-64}$ | 0.568222128508        | $2.38 \times 10^{-64}$ |
| 0.15 | 0.0101560362 | 2.388546340413        | $2.67 \times 10^{-63}$ | 0.646031450151        | $2.42 \times 10^{-63}$ |
| 0.25 | 0.0179723870 | 1.979779697106        | $8.58 \times 10^{-63}$ | 0.712164637068        | $7.87 \times 10^{-63}$ |
| 0.35 | 0.0268979678 | 1.780445201681        | $2.01 \times 10^{-62}$ | 0.783302404714        | $1.89 \times 10^{-62}$ |
| 0.45 | 0.0372955571 | 1.674431598537        | $4.16 \times 10^{-62}$ | 0.867134302856        | $4.08 \times 10^{-62}$ |
| 0.55 | 0.0497422622 | 1.627511940004        | $8.26 \times 10^{-62}$ | 0.972928209217        | $8.80 \times 10^{-62}$ |
| 0.65 | 0.0652421837 | 1.632385600040        | $1.69 \times 10^{-61}$ | 1.116892967890        | $2.06 \times 10^{-61}$ |
| 0.75 | 0.0857957337 | 1.702946079521        | $3.95 \times 10^{-61}$ | 1.334933069656        | $5.84 \times 10^{-61}$ |
| 0.85 | 0.1164390607 | 1.903257398990        | $1.46 \times 10^{-60}$ | 1.735363096266        | $2.40 \times 10^{-60}$ |
| 0.95 | 0.1793160070 | 2.658173924972        | $3.29 \times 10^{-59}$ | 2.995797394518        | $2.72 \times 10^{-59}$ |

The displayed block values are the  $c$ -recursive results; the adjacent columns give their relative differences from the independently generated  $h$ -recursive values. The maximum discrepancies are

$$\max_z \delta_e(z) = 3.2932 \times 10^{-59}, \quad \max_z \delta_o(z) = 2.7185 \times 10^{-59}, \quad (156)$$

both at  $z = 0.95$ , where the nome is largest.

At  $b = 1$ , separate Kac terms are resonant although their sum has a finite limit. We extracted the constant Laurent coefficient on a circle in  $t = \log b$ , using 32 equally spaced samples at 60-digit working precision. Repeating the extraction at radii 0.035 and 0.045 changes the  $c$ -recursive coefficients by at most  $9.53 \times 10^{-17}$  and the  $h$ -recursive coefficients by at most  $2.64 \times 10^{-43}$ . Coefficient by coefficient, the maximum direct  $c$ - versus  $h$ -recursive relative differences are  $7.47 \times 10^{-17}$  in the even family and  $9.52 \times 10^{-17}$  in the odd family. Thus the comparison supports the recursion both coefficientwise through the stated cutoff and after evaluating the complete truncated block.

An earlier version of this test converted the shifted internal weight in the large- $c$  seed to Python's binary64 complex type. Resonant terms amplified that loss before cancellation, producing artificial discrepancies of order  $10^{-3}$  and  $10^{-4}$  in the final retained coefficients. The corrected implementation preserves the `mpmath` numeric type throughout the seed; a regression test probes changes below binary64 resolution.

The standard-frame specialization is also enforced at the implementation level. The production multiprecision sphere block calls the principal-sheet, weight-only ordinary-edge scalar kernel in `Code/ns_recursion_recipe.py`, then supplies the canonical bra-incidence phase appropriate to this sphere ordering. Thus the module is not the full generic  $\widehat{\Sigma}$  operator of Eq. (38): canonical slot signs and graph Koszul transport remain with the graph caller. It also uses fresh principal square roots per call and is therefore not an analytic-continuation engine for complex-weight paths; such a caller must carry the continuous branch data required below Eq. (23).

As an independent nonrecursive check, a finite- $c$  PBW oracle constructs the NS Gram matrix and both descendant three-forms directly from the super-Virasoro commutators and Ward identities. At the generic point  $c = 14.19870372000744$ ,  $h = 0.73$ , and  $(h_1, h_2, h_3, h_4) = (0.37, 0.61, 0.48, 0.29)$ , it agrees with the sphere  $c$ -recursion through twice-level eight. For the unstarred block the maximum multiprecision discrepancy is  $3.26 \times 10^{-15}$ ; after extending the test to all four bottom/top choices at punctures two and three, the worst discrepancy is  $1.38 \times 10^{-14}$ . This range crosses both the first odd pole  $(3, 1)$  and the first even pole  $(2, 2)$ , without importing any Kac-residue ingredient into the PBW side.

The reproducible calculation and its complete coefficient ledger are

Code/compare\_ns\_sphere\_c\_h\_recursion.py,  
Machine Notes/c-Recursion/ns\_sphere\_c\_h\_order10.json.

### 6.1.2 Nonchiral sphere crossing from the direct $c$ -recursion

The preceding test compares two chiral recursion schemes at fixed internal momentum. We next assemble the nonchiral super-Liouville continuum and compare two different OPE channels. For four bottom NS primaries ordered as

$$\langle V_{P_4}(\infty) V_{P_3}(1) V_{P_2}(z, \bar{z}) V_{P_1}(0) \rangle, \quad (157)$$

write  $C_{ijP} = C(P_i, P_j, P)$  and  $\tilde{C}_{ijP} = \tilde{C}(P_i, P_j, P)$ . The direct-channel continuum decomposition is

$$G_{4321}^{(s)}(z, \bar{z}) = \int_0^\infty \frac{dP}{\pi} \left[ C_{12P} C_{34P} \mathcal{F}_e^{4321}(P; z) \mathcal{F}_e^{4321}(P; \bar{z}) + \tilde{C}_{12P} \tilde{C}_{34P} \mathcal{F}_o^{4321}(P; z) \mathcal{F}_o^{4321}(P; \bar{z}) \right]. \quad (158)$$

The  $t$ -channel is computed independently after the permutation  $(P_1, P_2, P_3, P_4) \mapsto (P_3, P_2, P_1, P_4)$ :

$$G_{4123}^{(t)}(1-z, 1-\bar{z}) = \int_0^\infty \frac{dP}{\pi} \left[ C_{32P} C_{14P} \mathcal{F}_e^{4123}(P; 1-z) \mathcal{F}_e^{4123}(P; 1-\bar{z}) + \tilde{C}_{32P} \tilde{C}_{14P} \mathcal{F}_o^{4123}(P; 1-z) \mathcal{F}_o^{4123}(P; 1-\bar{z}) \right]. \quad (159)$$

No equality between Eqs. (158) and (159) is imposed numerically.

Each chiral block is evaluated from the functional recursion (149) itself. At every leaf we retain the exact hypergeometric global  $\mathfrak{osp}(1|2)$  block of Eq. (206); there is no local- $z$  or elliptic- $q$  series

cutoff. Recursion order  $N$  retains every nested Kac-residue path whose accumulated physical null level is at most  $N$ . The implementation constructs the residue tree once per internal momentum and evaluates its hypergeometric leaves on the full  $z$ -grid.

At the self-dual point we use

$$c = \frac{27}{2}, \quad b = 1, \quad (P_1, P_2, P_3, P_4) = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{3}{5}\right). \quad (160)$$

The order-twelve run uses 80-digit block arithmetic, 35-digit structure constants,  $P \leq 5$ , and a 40-point Gauss–Legendre quadrature. For

$$\delta_{s/t}(z) = \frac{|G_{4321}^{(s)}(z) - G_{4123}^{(t)}(1-z)|}{\max(|G_{4321}^{(s)}(z)|, |G_{4123}^{(t)}(1-z)|)}, \quad (161)$$

the independently computed values are

| $z$  | $G_{4321}^{(s)}(z)$ | $G_{4123}^{(t)}(1-z)$ | $\delta_{s/t}(z)$       |
|------|---------------------|-----------------------|-------------------------|
| 0.30 | 0.250435355646635   | 0.250435348477313     | $2.863 \times 10^{-8}$  |
| 0.35 | 0.228247438016132   | 0.228247436965260     | $4.604 \times 10^{-9}$  |
| 0.40 | 0.213525561198958   | 0.213525561041801     | $7.360 \times 10^{-10}$ |
| 0.45 | 0.204170165991172   | 0.204170165968469     | $1.112 \times 10^{-10}$ |
| 0.50 | 0.199078377621857   | 0.199078377622339     | $2.419 \times 10^{-12}$ |
| 0.55 | 0.197758604465580   | 0.197758604490486     | $1.259 \times 10^{-10}$ |
| 0.60 | 0.200188174928437   | 0.200188175091731     | $8.157 \times 10^{-10}$ |
| 0.65 | 0.206826227176964   | 0.206826228216805     | $5.028 \times 10^{-9}$  |
| 0.70 | 0.218798349671143   | 0.218798356420083     | $3.085 \times 10^{-8}$  |

Thus the observed order-twelve crossing residual satisfies

$$\max_{0.30 \leq z \leq 0.70} \delta_{s/t}(z) = 3.0846 \times 10^{-8}. \quad (162)$$

Figure 1 displays both channel sums and their signed relative difference; the two channel curves have not been averaged or otherwise forced to coincide.

The accuracy claim requires a qualification. At the same 40 momentum nodes, the maximum order-ten crossing residual is  $3.7697 \times 10^{-7}$ , while the maximum order-ten to order-twelve drift of either individual channel is  $3.4612 \times 10^{-7}$ . The number in Eq. (162) is therefore an observed order-twelve residual, not a certified  $3 \times 10^{-8}$  error bound for the infinite-recursion correlator. Increasing the recursion order is postponed. Moreover, crossing cannot diagnose a systematic error common to both channel implementations. The independent  $c$ - versus  $h$ -recursion comparison above tests the chiral kernel, but the nonchiral structure-constant assembly and continuum normalization remain shared inputs in this crossing test.

The reproducible implementation, raw ledger, and plot are

```
Code/stress_ns_crossing.py,
DataSet/ns_liouville_crossing_direct_c_recursion_order10.json,
DataSet/ns_liouville_crossing_direct_c_recursion_order12.json,
DataSet/ns_liouville_crossing_direct_c_recursion.svg.
```

## 6.2 Torus one-point block

The torus one-point block provides a second independent comparison with a known internal-weight recursion. Closing the single NS edge into a handle, Eq. (2) specializes to

$$\mathcal{F}_{1,1}(c, h, d; q, \xi) = \mathcal{U}_{1,1}(h, d; q, \xi)$$

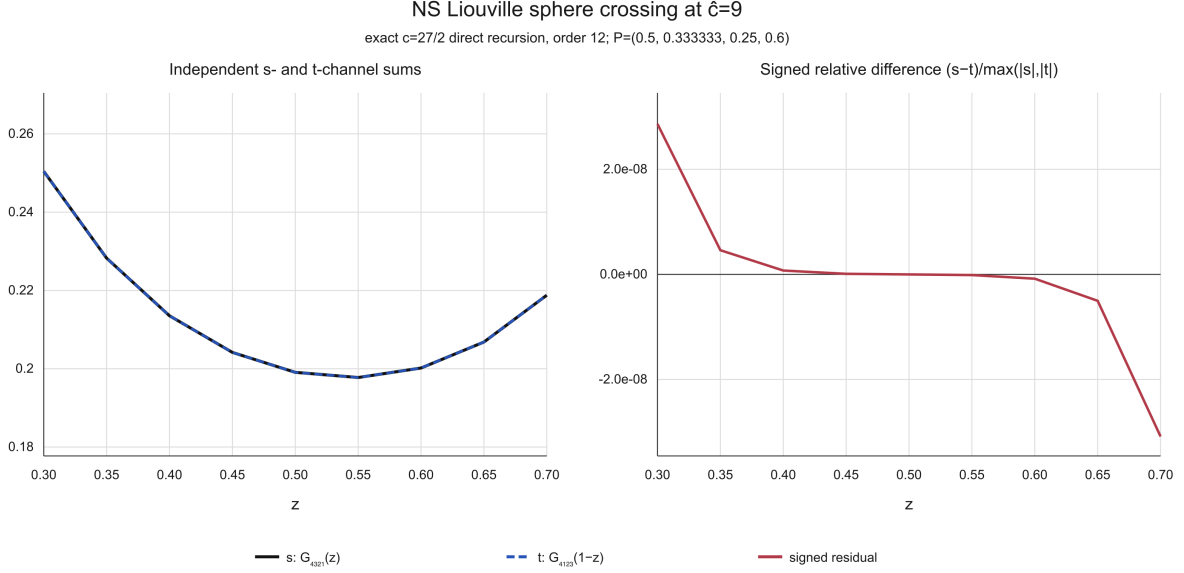


Figure 1: Direct  $c$ -recursion evaluation of the same nonchiral NS sphere four-point function in the  $s$ - and  $t$ -channels at  $\hat{c} = 9$ . The right panel shows  $(G^{(s)} - G^{(t)}) / \max(|G^{(s)}|, |G^{(t)}|)$ , without assuming crossing.

$$+ \sum_{\substack{r \geq 2, s \geq 1 \\ r+s \in 2\mathbb{Z}}} \frac{q^{rs/2} \xi^{rs \bmod 2} \mathcal{R}_{r,s}^{\text{tor}}(h, d)}{c - c_{r,s}(h)} \mathcal{F}_{1,1} \left( c_{r,s}(h), h + \frac{rs}{2}, d; q, \xi \right). \quad (163)$$

The regular term is the product of the non-global NS vacuum character (109) and the global torus block (207). The residue is the single sequential toric null matrix element

$$\mathcal{R}_{r,s}^{\text{tor}}(h, d) = J_{r,s}(h) A_{r,s}(h) S_{rs}^0 P_{r,s}^{\pi_{r,s}}(d, h + N_{r,s}; h) P_{r,s}^0(d, h; h), \quad (164)$$

which is Eq. (45) in the standard torus rephasing. This fixed-weight  $c$ -recursion was compared with the toric  $h$ -recursion of Ref. [10].

The comparison concerns the normalized chiral conformal block only. We set

$$\rho(\nu_h, \nu_d, \nu_h) = 1. \quad (165)$$

No super-Liouville structure constant, reflection amplitude, spectral measure, or momentum integral is used. The fusion polynomials in Eq. (163) are representation-theoretic null-vector matrix elements, not dynamical three-point coefficients.

We again used the self-dual point

$$\hat{c} = 9, \quad c = \frac{27}{2}, \quad b = 1, \quad P_{\text{int}} = \frac{7}{10}, \quad h = \frac{149}{200}, \quad P_{\text{ext}} = \frac{1}{2}, \quad d = \frac{5}{8}. \quad (166)$$

All integer and half-integer powers through  $q^{10}$  were retained. In contrast with the sphere plumbing tree, the two values of  $\xi$  are now the two invariant temporal spin holonomies around the closed torus cycle:  $\xi = +1$  gives the ordinary NS trace and  $\xi = -1$  inserts  $(-1)^F$ .

For the relative discrepancy

$$\delta_\xi(q) = \frac{|\mathcal{F}_{1,1}^{(c)}(q, \xi) - \mathcal{F}_{1,1}^{(h)}(q, \xi)|}{\max(|\mathcal{F}_{1,1}^{(c)}(q, \xi)|, |\mathcal{F}_{1,1}^{(h)}(q, \xi)|)}, \quad (167)$$

the ten-point result is

| $q$  | $\mathcal{F}_{\xi=+1}^{(c)}$ | $\delta_+$             | $\mathcal{F}_{\xi=-1}^{(c)}$ | $\delta_-$             |
|------|------------------------------|------------------------|------------------------------|------------------------|
| 0.02 | 0.540548101410               | $3.56 \times 10^{-64}$ | 0.456229368384               | $3.36 \times 10^{-64}$ |
| 0.04 | 0.647543530569               | $1.07 \times 10^{-63}$ | 0.505419160552               | $1.54 \times 10^{-61}$ |
| 0.06 | 0.732306455976               | $2.10 \times 10^{-63}$ | 0.535126888371               | $2.04 \times 10^{-63}$ |
| 0.08 | 0.809428769226               | $3.44 \times 10^{-63}$ | 0.556552964433               | $3.54 \times 10^{-63}$ |
| 0.10 | 0.884278365341               | $5.11 \times 10^{-63}$ | 0.573367584001               | $5.69 \times 10^{-63}$ |
| 0.12 | 0.959753226892               | $1.62 \times 10^{-61}$ | 0.587248485524               | $8.75 \times 10^{-63}$ |
| 0.14 | 1.037845659532               | $9.62 \times 10^{-63}$ | 0.599106998669               | $1.31 \times 10^{-62}$ |
| 0.16 | 1.120192330357               | $1.39 \times 10^{-61}$ | 0.609494891429               | $3.83 \times 10^{-61}$ |
| 0.18 | 1.208326107532               | $1.71 \times 10^{-62}$ | 0.618772556207               | $5.04 \times 10^{-61}$ |
| 0.20 | 1.303821014733               | $2.37 \times 10^{-62}$ | 0.627189301630               | $6.21 \times 10^{-61}$ |

Thus

$$\max_q \delta_+(q) = 1.6226 \times 10^{-61}, \quad \max_q \delta_-(q) = 6.2132 \times 10^{-61}. \quad (168)$$

The coefficient-by-coefficient comparison is equally sharp:

$$\max_{0 \leq n \leq 20} \frac{|F_{n/2}^{(c)} - F_{n/2}^{(h)}|}{\max(|F_{n/2}^{(c)}|, |F_{n/2}^{(h)}|)} = 5.186 \times 10^{-58}. \quad (169)$$

The self-dual finite part was extracted at 60-digit working precision from 32 samples on each of the radii 0.035 and 0.045. Changing the radius alters either recursion by at most  $7.35 \times 10^{-51}$ . At the generic nonresonant point  $b = 1.27$ , the two recursions agree through  $q^6$  to  $1.12 \times 10^{-60}$ . The production standard torus block uses the principal-sheet incidence-ordered self-loop scalar kernel in `Code/ns_recursion_recipe.py`. Independently, direct Gram-matrix and Ward-identity sewing at  $c = 14.19870372000744$ ,  $h = 0.73$ , and  $d = 0.62$  agrees through twice-level eight to  $2.64 \times 10^{-13}$ . Thus this check also crosses the odd (3, 1) and even (2, 2) nulls and tests the intermediate sector toggle, shifted second incidence, and toric sign without reusing them on the PBW side. This complete block comparison is for  $\alpha = 0$ . A separate symmetric Laurent extraction from the direct PBW trace checks the local self-loop residues for both  $\alpha = 0, 1$  at (3, 1) and (2, 2), with maximum discrepancy  $2.18 \times 10^{-12}$ . It does not construct the full  $\alpha = 1$  top-component regular block. The order-ten result is therefore an agreement of the normalized conformal block, including both torus spin holonomies, with no theory-dependent OPE data.

The reproducible implementation and numerical ledger are

`Code/compare_ns_torus_c_h_recursion.py`,  
`Machine Notes/c-Recursion/ns_torus_c_h_order10.json`.

### 6.2.1 Nonchiral NS–NS modularity from the direct $c$ -recursion

The preceding comparison validates the chiral recursion coefficient by coefficient. We now test a genuinely different statement: modular covariance of the assembled nonchiral super-Liouville continuum. A fixed-momentum block is not modular covariant by itself, because the modular kernel mixes internal representations. For the ordinary NS trace, which is fixed as a spin structure by  $S$ , define

$$\mathcal{G}_{\text{NS}}(\tau) = \int_0^\infty \frac{dP}{\pi} C(P, P_{\text{ext}}, P) \left| q^{h(P)-c/24} \mathcal{F}_{1,1}(c, h(P), d; q, +1) \right|^2, \quad q = e^{2\pi i \tau}. \quad (170)$$

The two frames were evaluated independently and compared with

$$\mathcal{G}_{\text{NS}}(-1/\tau) = |\tau|^{2d} \mathcal{G}_{\text{NS}}(\tau). \quad (171)$$

No equality between the two numerical integrals was imposed.

As in the sphere test, the block is evaluated from the functional recursion (163), rather than from a truncated coefficient series. Every terminal node retains the exact large- $c$  seed

$$\mathcal{U}_{1,1}(h, d; q, \xi) = \mathcal{V}_{\text{NS}}^{(\infty)}(q, \xi) \mathcal{G}_{1,1}^{\text{osp}(1|2)}(h, d; q, \xi), \quad (172)$$

where the non-global vacuum factor is the convergent product

$$\mathcal{V}_{\text{NS}}^{(\infty)}(q, \xi) = \frac{(-\xi q^{3/2}; q)_{\infty}}{(q^2; q)_{\infty}}, \quad (173)$$

and the global trace has the closed form

$$\begin{aligned} \mathcal{G}_{1,1}^{\text{osp}(1|2)}(h, d; q, \xi) &= (1 - q)^{-d} {}_2F_1(2h - d, 1 - d; 2h; q) \\ &\quad + \xi q^{1/2} \frac{2h - d}{2h} (1 - q)^{-d} {}_2F_1(2h + 1 - d, 1 - d; 2h + 1; q). \end{aligned} \quad (174)$$

Expanding Eqs. (172)–(174) reproduces the regular coefficients used in the independent  $c$ - versus  $h$ -recursion check. Recursion order  $N$  retains every nested Kac-residue path whose accumulated twice-level is at most  $N$ ; it does not truncate a local- $q$  or elliptic- $q$  series.

It is important that a modular test force at least one plumbing frame out of the very-small- $q$  regime. Our first scan,  $\tau = 0.2 + iy$  with  $0.65 \leq y \leq 1.20$ , did not do this: throughout that scan

$$|q| \leq 1.684 \times 10^{-2}, \quad |\tilde{q}| = |e^{-2\pi i/\tau}| \leq 6.131 \times 10^{-3}. \quad (175)$$

The order-fourteen residual  $2.23 \times 10^{-14}$  found there is therefore retained only as a small- $q$  implementation calibration. It is not quoted as the nontrivial modularity result.

For the hard audit we instead choose

$$\hat{c} = 9, \quad b = 1, \quad P_{\text{ext}} = 0.33, \quad d = 0.55445, \quad \tau = \frac{i}{4}, \quad -\frac{1}{\tau} = 4i, \quad (176)$$

so that

$$|q| = e^{-\pi/2} = 0.2078795764, \quad |\tilde{q}| = e^{-8\pi} = 1.216155671 \times 10^{-11}. \quad (177)$$

The continuum is integrated through  $P = 5.5$  with a 96-point Gauss–Legendre rule. Blocks use 100–110-digit arithmetic and the structure constants use 45-digit arithmetic. The modular factor is applied only after the two continuum sums have been completed: it is absent from the block recursion, structure constant, and quadrature accumulation in either frame. The recursion-order audit is

| $N$ | $\mathcal{G}_{\text{NS}}(i/4)$ | $4^{2d}\mathcal{G}_{\text{NS}}(4i)$ | $\delta_S$              |
|-----|--------------------------------|-------------------------------------|-------------------------|
| 12  | 0.419420508925143              | 0.419421352297236                   | $2.0108 \times 10^{-6}$ |
| 14  | 0.419422873756867              | 0.419421352297236                   | $3.6275 \times 10^{-6}$ |
| 16  | 0.419421128663541              | 0.419421352297236                   | $5.3320 \times 10^{-7}$ |
| 18  | 0.419421333942803              | 0.419421352297236                   | $4.3761 \times 10^{-8}$ |
| 20  | 0.419421361508904              | 0.419421352297236                   | $2.1963 \times 10^{-8}$ |

Here

$$\delta_S(\tau) = \frac{|\mathcal{G}_{\text{NS}}(\tau) - |\tau|^{-2d}\mathcal{G}_{\text{NS}}(-1/\tau)|}{\max(|\mathcal{G}_{\text{NS}}(\tau)|, ||\tau|^{-2d}\mathcal{G}_{\text{NS}}(-1/\tau)|)}. \quad (178)$$

Thus the observed order-twenty residual is

$$\delta_S(i/4) = 2.1963 \times 10^{-8}. \quad (179)$$



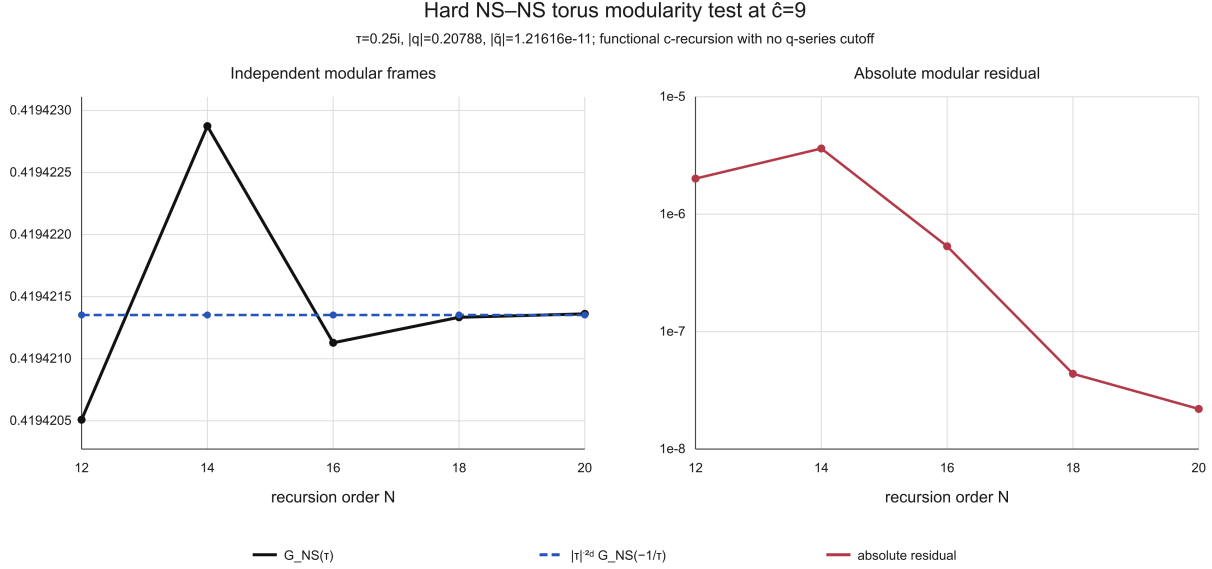


Figure 2: Hard-point recursion-order convergence. Solid black markers show  $\mathcal{G}_{\text{NS}}(i/4)$ , while dashed blue markers show the independently evaluated  $4^{2d}\mathcal{G}_{\text{NS}}(4i)$ . Since  $|\tilde{q}| = 1.22 \times 10^{-11}$ , the dashed markers coincide; their common horizontal value is output, not an imposed target. The right panel shows the absolute modular residual.

The residual is an observed consistency check, not a certified error bound for the infinite recursion. The largest order-eighteen to order-twenty drift of either individual frame is  $6.5724 \times 10^{-8}$ , larger than Eq. (179); the nonmonotonic order-twelve to order-fourteen step also makes clear that the last residual alone must not be used as an error estimate. By contrast, replacing  $(P_{\text{max}}, n_P) = (5.0, 64)$  with  $(5.5, 96)$  changes either frame at orders twelve and fourteen by at most  $1.31 \times 10^{-14}$ . The hard-point uncertainty is therefore controlled by recursion order, not by the displayed quadrature or momentum tail. Modularity still cannot expose a systematic error common to the two frames; the structure constants and spectral normalization remain common inputs.

The reproducible implementation, numerical ledger, and plot are

```
Code/stress_ns_torus_modularity_c_recursion.py,  
Data Set/ns_torus_modularity_direct_c_recursion_hard_tau025.json,  
Data Set/ns_torus_modularity_direct_c_recursion_hard_tau025.svg.
```

The earlier calibration is preserved separately as `Data Set/ns_torus_modularity_direct_c_recursion_small_q_order14.json`.

### 6.3 Genus-two vacuum graphs

Let  $h_1, h_2, h_3$  and  $q_1, q_2, q_3$  label the three edges. A pole on edge  $i$  gives

$$\widehat{\mathcal{R}}_{i;r,s} \Big|_{\mathbf{p}} = J_{r,s}(h_i) A_{r,s}(h_i) \widehat{\Sigma}_{r,s}^{(L,i)} \Big|_{\mathbf{p}_L}(h_j, h_k; h_i) \widehat{\Sigma}_{r,s}^{(R,i)} \Big|_{\mathbf{p}_R}(h_j, h_k; h_i), \quad \{i, j, k\} = \{1, 2, 3\}. \quad (180)$$

Equation (180) applies to the theta graph, whose three edges all have distinct endpoints. In the glasses graph the separating bridge also uses Eq. (43), whereas each of the two handle edges is a

self-loop and must use Eq. (45). This is exactly the sequential shifted-weight prescription tested against the torus block above. The glasses implementation also retains the toric reflection sign

$$S_{rs}^\alpha = (-1)^{\alpha rs}. \quad (181)$$

This factor is invisible in an  $\alpha = 0$  regression. The production check therefore compares both  $\alpha = 0, 1$  sectors with the independent torus formula for the odd  $(3, 1)$  and even  $(2, 2)$  nulls. At  $(h_L, h_B, \alpha) = (0.73, 0.62, 1)$ , the corrected  $(3, 1)$  residue is  $+0.059809637121$ ; omitting Eq. (181) gives the negative of this value, while the even-null residue is unchanged. Writing twice-levels as  $\ell, v, r$ , the regular term is

$$U_{\Theta, \ell}^{\alpha_L, \alpha_R} = \sum_{v+r=\ell} (-1)^{B_\Theta(\bar{v}, \bar{r})} V_{\Theta, v}^{(\infty)} G_{\Theta, r}^{\text{osp}(1|2), \alpha_L, \alpha_R}, \quad (182)$$

Here  $B_\Theta$  is the six-term polynomial (69), the global coefficient is Eq. (91), and the vacuum coefficients are obtained by expanding Eq. (107) with the generators (198). The independently sewn coefficients through total level six are displayed in Eq. (201). Deleting the displayed  $(-1)^{B_\Theta}$  gives the wrong answer beginning at total level  $7/2$ ; using the correct primitive product gives the same vacuum coefficients as direct sewing.

### 6.3.1 Function-level genus-two global resummation

The global factor in Eq. (182) must be evaluated at every terminal node of the functional  $c$ -recursion, including nodes with shifted internal weights. A fixed cutoff on the three global occupations is therefore both expensive and logically independent of the recursion order. In the two genus-two channels used in the locality test, the occupation sums can instead be reorganized into Gauss hypergeometric functions. We keep the plumbing square-root lifts  $\xi_e = \pm 1$  explicit and use the norm (76) throughout.

**Glasses channel.** Write the handle and bridge data as

$$(h_L, q_L, \xi_L), \quad (h_R, q_R, \xi_R), \quad (d, q_B, \xi_B), \quad (183)$$

and fix the bridge fermion parity to  $\alpha = \epsilon_B \in \{0, 1\}$ . Since a handle state occurs twice at its trinion, both vertex parities equal  $\alpha$ . In the ordered sewing frame the global orientation character is

$$\Omega_{\text{gl}} = (-1)^{\alpha(\epsilon_L + \epsilon_R)}. \quad (184)$$

The direct global contraction is therefore

$$\begin{aligned} \mathcal{G}_{\text{gl}}^{(\alpha)} = & \sum_{\substack{n_L, n_R, n_B \geq 0 \\ \epsilon_L, \epsilon_R \in \{0, 1\}}} (-1)^{\alpha(\epsilon_L + \epsilon_R)} \prod_{e=L, R, B} \frac{q_e^{n_e + \epsilon_e/2} \xi_e^{\epsilon_e}}{n_e! (2h_e)_{n_e + \epsilon_e}} \\ & \times T_{n_L, n_B, n_L}^{\epsilon_L \alpha \epsilon_L} (h_L, d, h_L) T_{n_R, n_B, n_R}^{\epsilon_R \alpha \epsilon_R} (h_R, d, h_R), \quad \epsilon_B = \alpha, \end{aligned} \quad (185)$$

where  $h_B = d$ . Equation (88) gives

$$T_{n, n_B, n}^{\epsilon \alpha \epsilon} (h, d, h) = \left( -d - \frac{\alpha}{2} \right)^{\frac{n_B}{2}} t_{n, n}^{\epsilon \alpha \epsilon} (h, d, h). \quad (186)$$

Thus the bridge occupation separates from both handle traces. Define

$$D_\alpha := d + \frac{\alpha}{2}, \quad C_0(h, d) := 2h - d, \quad C_1(h, d) := 2h + d - \frac{1}{2}. \quad (187)$$

The bridge factor is

$$B_\alpha(q, \xi; d) = \frac{(\xi\sqrt{q})^\alpha}{(2d)_\alpha} {}_2F_1(D_\alpha, D_\alpha; 2d + \alpha; q), \quad (188)$$

while one handle contributes

$$H_\alpha(q, \xi; h, d) = (1 - q)^{-D_\alpha} \left[ {}_2F_1(2h - D_\alpha, 1 - D_\alpha; 2h; q) + (-1)^\alpha \xi \sqrt{q} \frac{C_\alpha(h, d)}{2h} {}_2F_1(2h + 1 - D_\alpha, 1 - D_\alpha; 2h + 1; q) \right]. \quad (189)$$

The factors  $(-1)^\alpha$  multiplying both handle lifts absorb Eq. (184). Consequently the complete glasses global block is the function-level product

$$\mathcal{G}_{\text{gl}}^{(\alpha)} = B_\alpha(q_B, \xi_B; d) H_\alpha(q_L, \xi_L; h_L, d) H_\alpha(q_R, \xi_R; h_R, d). \quad (190)$$

This identity removes all three occupation cutoffs. At  $q_B = 0$ , the odd sector vanishes and the even sector reduces to the product of the two independently known torus global blocks (174).

**Theta channel.** Keep the geometric edge order  $(0, 1, \infty)$ , but rotate the trinion to the CCY tensor order  $(\infty, 1, 0)$ . For fixed endpoint states set

$$X = h_\infty - h_1 - h_0 + \frac{\epsilon_\infty - \epsilon_1 - \epsilon_0}{2} + n_\infty - n_0. \quad (191)$$

Translation covariance, Eq. (88), then isolates the complete middle chain:

$$T_{n_\infty, n_1, n_0}^{\epsilon_\infty \epsilon_1 \epsilon_0}(h_\infty, h_1, h_0) = X_{n_1}^{n_1} t_{n_\infty, n_0}^{\epsilon_\infty \epsilon_1 \epsilon_0}(h_\infty, h_1, h_0). \quad (192)$$

Using  $X^n = (-1)^n (-X)_n$ , the entire occupation sum on the middle edge is

$$\begin{aligned} \sum_{n_1=0}^{\infty} \frac{q_1^{n_1 + \epsilon_1/2} \xi_1^{\epsilon_1}}{n_1! (2h_1)_{n_1 + \epsilon_1}} (X^{n_1})^2 \\ = \frac{(\xi_1 \sqrt{q_1})^{\epsilon_1}}{(2h_1)_{\epsilon_1}} {}_2F_1(-X, -X; 2h_1 + \epsilon_1; q_1). \end{aligned} \quad (193)$$

For a fixed theta sector  $\kappa$ , the remaining representation is the two-endpoint sum

$$\begin{aligned} \mathcal{G}_\Theta^{(\kappa)} = \sum_{\substack{n_0, n_\infty \geq 0 \\ \epsilon_0, \epsilon_1, \epsilon_\infty \in \{0, 1\} \\ \epsilon_0 + \epsilon_1 + \epsilon_\infty \equiv \kappa \pmod{2}}} \Omega_\Theta(\epsilon) \prod_{e=0, \infty} \frac{q_e^{n_e + \epsilon_e/2} \xi_e^{\epsilon_e}}{n_e! (2h_e)_{n_e + \epsilon_e}} \\ \times \left[ t_{n_\infty, n_0}^{\epsilon_\infty \epsilon_1 \epsilon_0}(h_\infty, h_1, h_0) \right]^2 \frac{(\xi_1 \sqrt{q_1})^{\epsilon_1}}{(2h_1)_{\epsilon_1}} {}_2F_1(-X, -X; 2h_1 + \epsilon_1; q_1), \end{aligned} \quad (194)$$

where  $\Omega_\Theta$  is the full orientation character (196). Unlike the glasses graph, the two endpoint sums do not factorize into independent one-variable traces. They are kept as an adaptive numerical representation after the non-small middle coordinate has been summed exactly. At the five certified overlap points,

$$2.44 \times 10^{-9} \leq |q_0| \leq 4.44 \times 10^{-9}, \quad 2.84 \times 10^{-9} \leq |q_\infty| \leq 4.92 \times 10^{-9}, \quad 0.148 \leq |q_1| \leq 0.178. \quad (195)$$

Thus Eq. (193) resums precisely the edge for which a direct occupation cutoff is potentially dangerous. The endpoint shell is an independently monitored numerical tolerance, not a truncation of the Zamolodchikov recursion.

Both identities were checked without assuming channel agreement. For the glasses formula, direct sums in both bridge parities and with complex plumbing coordinates agree at the  $10^{-15}$  level whenever the direct sum converges. For the theta formula, twenty-four randomized tests with all three  $|q_e|$  between 0.025 and 0.13 gave a worst relative discrepancy  $1.55 \times 10^{-13}$ , consistent with the  $2 \times 10^{-12}$  tolerance of the independent three-dimensional reference sum. At the production theta point with the largest old last-shell diagnostic, the monitored shell changed from  $1.65 \times 10^{-9}$  to  $9.14 \times 10^{-36}$ ; the corresponding integrated node changed by  $1.22 \times 10^{-10}$ . At the worst old glasses node the last shell was  $3.27 \times 10^{-2}$ , while the resummed integrated node changed by  $6.80 \times 10^{-6}$ . These are seed audits only and do not assert locality or equality of the final theta and glasses integrals.

The function-level implementations and their direct-sum regression tests are

$$\text{Code/ns\_genus2\_partition.py}, \quad \text{Code/test\_ns\_genus2\_partition.py}.$$

### 6.3.2 Theta-chart vacuum product

For a concrete genus-two check, use the theta plumbing coordinates  $q_0, q_1, q_\infty$ . For edge fermion parities  $\epsilon = (\epsilon_0, \epsilon_1, \epsilon_\infty)$ , the fixed half-edge ordering used below has

$$\Omega_\Theta(\epsilon) = (-1)^{\epsilon_0 \epsilon_1 + \epsilon_0 \epsilon_\infty + \epsilon_1 \epsilon_\infty + \epsilon_\infty}. \quad (196)$$

This is now derived rather than entered by hand. In vertex-slot order  $(0_L, 1_L, \infty_L, 0_R, 1_R, \infty_R)$ , the half-edge frame signs are

$$((-1)^{f_h})_{h \in H_V} = (+, +, +, +, +, -), \quad (t_0, t_1, t_\infty) = (0, 0, 0). \quad (197)$$

The contraction order  $(0_L, 0_R, 1_L, 1_R, \infty_L, \infty_R)$  supplies the three quadratic inversion terms, while Eq. (59) gives  $(\beta_0, \beta_1, \beta_\infty) = (0, 0, 1)$ . Their sum is exactly Eq. (196). Consequently an odd null on edge  $e$  carries the coefficient transport  $\Omega_\Theta(\epsilon)/\Omega_\Theta(\epsilon \oplus e_e)$ . In the CCY marking, the two projective generators are

$$\gamma_A(z) = q_0 q_\infty z, \quad \gamma_B(z) = \frac{(1 - q_1)z - q_0^{-1}}{z - q_0^{-1}}. \quad (198)$$

The second map depends on  $q_0$ , not  $q_\infty$ . Replacing one by the other preserves the three leading pinching monomials but fails beginning at the next order. If  $q_e = tx_e$ , the three shortest lifted primitive classes  $A, B, AB$  have

$$u_A = \xi_0 \xi_\infty (q_0 q_\infty)^{1/2}, \\ u_B = -\xi_0 \xi_1 (q_0 q_1)^{1/2} (1 + O(t)),$$

$$u_{AB} = \xi_1 \xi_\infty (q_1 q_\infty)^{1/2} (1 + O(t)). \quad (199)$$

The middle minus is generated by the determinant-one lift of  $\gamma_B$ ; it is not an independently assigned sign. The three shortest supercurrent links therefore give

$$\mathcal{V}_{\Theta, \text{NS}}^{(\infty)} = 1 + \xi_0 \xi_\infty (q_0 q_\infty)^{3/2} + \xi_1 \xi_\infty (q_1 q_\infty)^{3/2} - \xi_0 \xi_1 (q_0 q_1)^{3/2} + O(t^4). \quad (200)$$

Bosonic stress-tensor links begin at order  $t^4$ . The companion Python implementation computes the exact Schottky multipliers rather than using these leading expansions. Expanding that product gives the ordered CCY-frame vacuum quotient. Put  $x_e = \xi_e q_e^{1/2}$ . Direct vacuum Gram-matrix sewing and the  $c \rightarrow \infty$  limit give

$$\mathcal{V}_{\Theta, \text{pl}}^{(\infty)} = 1 + V_6 + V_8 + V_{10} + V_{12} + O(|\mathbf{x}|^{13}), \quad (201)$$

where the subscript is total twice-level and

$$\begin{aligned} V_6 &= x_1^3 x_\infty^3 + x_0^3 x_\infty^3 - x_0^3 x_1^3, \\ V_8 &= 3x_1^3 x_\infty^5 + x_1^4 x_\infty^4 + 3x_1^5 x_\infty^3 + x_0^4 x_\infty^4 + x_0^4 x_1^4 - 3x_0^5 x_1^3, \\ V_{10} &= 6x_1^3 x_\infty^7 + 4x_1^4 x_\infty^6 + 16x_1^5 x_\infty^5 + 4x_1^6 x_\infty^4 + 6x_1^7 x_\infty^3 \\ &\quad + x_0^5 x_\infty^5 - x_0^5 x_1^5 + 4x_0^6 x_1^4 - 6x_0^7 x_1^3, \\ V_{12} &= 10x_1^3 x_\infty^9 + 10x_1^4 x_\infty^8 + 50x_1^5 x_\infty^7 + 25x_1^6 x_\infty^6 + 50x_1^7 x_\infty^5 \\ &\quad + 10x_1^8 x_\infty^4 + 10x_1^9 x_\infty^3 + x_0^6 x_\infty^6 + x_0^6 x_1^6 - 8x_0^7 x_1^5 + 10x_0^8 x_1^4 - 10x_0^9 x_1^3. \end{aligned} \quad (202)$$

The asymmetry is not a typo: this is the fixed ordered  $(0, 1, \infty)$  trinion/BPZ frame. Equation (201) is independently regenerated in Python by removing  $G_{-1/2}|0\rangle$  and  $L_{-1}|0\rangle$ , sewing the resulting vacuum quotient, and Richardson-extrapolating in  $1/c$ . It agrees with the corrected primitive product through every displayed coefficient.

## 6.4 Global-block specializations

### 6.4.1 CCY-style sphere six-point block

The closest comparison with CCY Section 4.3 is the sphere channel with three internal weights  $h_i$ , three plumbing parameters  $q_i$ , and external pairs  $(d_1, d_2)$ ,  $(d_3, d_4)$ ,  $(d_5, d_6)$ . For bottom external components, define the three endpoint factors

$$E_i(n, \epsilon) = \left( h_i + d_{2i} - d_{2i-1} + \frac{\epsilon}{2} \right)_n. \quad (203)$$

Let  $\alpha_0$  be the central-trinion sector and  $\alpha_i$  the three outer sectors. Then

$$\begin{aligned} \mathcal{G}_{6, \text{tri}}^{\text{osp}(1|2), \alpha} &= \sum_{\substack{n_i \geq 0 \\ \epsilon_i \in \{0, 1\}}} \prod_{i=1}^3 \frac{q_i^{n_i + \epsilon_i/2} \xi_i^{\epsilon_i} E_i(n_i, \epsilon_i)}{n_i! (2h_i)_{n_i + \epsilon_i}} \\ &\quad \times \left[ \prod_{i=1}^3 \delta_{\epsilon_i, \alpha_i} \right] \delta_{\epsilon_1 + \epsilon_2 + \epsilon_3 \equiv \alpha_0 \pmod{2}} T_{n_1, n_2, n_3}^{\epsilon_1 \epsilon_2 \epsilon_3} (h_1, h_2, h_3). \end{aligned} \quad (204)$$

Up to the displayed parity projectors and the replacement of the central  $sl(2)$  tensor by Eq. (88), this is exactly CCY's Eq. (4.23). In particular, setting all  $\epsilon_i = 0$  reduces the central tensor to the CCY three-chain polynomial.

### 6.4.2 Sphere four-point global block

For four bottom components at  $0, z, 1, \infty$ , fixed even or odd chiral vertex sectors select  $\epsilon = 0$  or  $1$ . Equation (91) gives

$$\mathcal{G}_h^{\text{osp}(1|2),(\epsilon)}(z) = \sum_{n=0}^{\infty} \frac{(h + \epsilon/2 + h_2 - h_1)_n (h + \epsilon/2 + h_3 - h_4)_n}{n! (2h)_{n+\epsilon}} z^{n+\epsilon/2}. \quad (205)$$

Equivalently,

$$\begin{aligned} \mathcal{G}_h^{\text{osp}(1|2),(0)}(z) &= {}_2F_1(h + h_2 - h_1, h + h_3 - h_4; 2h; z), \\ \mathcal{G}_h^{\text{osp}(1|2),(1)}(z) &= \frac{z^{1/2}}{2h} {}_2F_1\left(h + h_2 - h_1 + \frac{1}{2}, h + h_3 - h_4 + \frac{1}{2}; 2h + 1; z\right). \end{aligned} \quad (206)$$

Their lift-refined direct sum is  $\mathcal{G}_h^{\text{osp}(1|2),(0)} + \xi \mathcal{G}_h^{\text{osp}(1|2),(1)}$ . It is the light block in the component-normalized convention of Ref. [2], whereas fixed parity-homogeneous three-forms use one summand. This is the four-point instance of Eq. (78).

### 6.4.3 Torus one-point global block

For one internal edge closed into a handle and a bottom primary insertion of weight  $d$ , both ends of the edge carry the same  $\epsilon$ , so the even trinion structure admits both integer and half-integer levels:

$$\mathcal{G}_{1,1}^{\text{osp}(1|2),(0)}(h, d; q, \xi) = \sum_{n \geq 0} \sum_{\epsilon=0}^1 \frac{q^{n+\epsilon/2} \xi^\epsilon T_{n,0,n}^{\epsilon,0,\epsilon}(h, d, h)}{n! (2h)_{n+\epsilon}}. \quad (207)$$

The first terms are

$$\mathcal{G}_{1,1}^{\text{osp}(1|2),(0)} = 1 + \xi q^{1/2} \frac{2h - d}{2h} + q \frac{2h + d(d-1)}{2h} + O(q^{3/2}), \quad (208)$$

in agreement with the direct super-Virasoro computation of Ref. [2].

## 6.5 Coefficient implementation

At a fixed cutoff, store the vacuum and global series as dictionaries indexed by twice-level vectors,

$$V = \sum_v V_v \mathbf{q}^{v/2}, \quad G = \sum_r G_r \mathbf{q}^{r/2}. \quad (209)$$

Ordinary multiplication of the already oriented scalar series would omit the polarized sign. The explicit theta-plumbing loop is

```
for vacuum_level, vacuum_coefficient in vacuum.items():
    global_level = level - vacuum_level
    if min(global_level) < 0:
        continue
    vacuum_parity = tuple(n % 2 for n in vacuum_level)
    global_parity = tuple(n % 2 for n in global_level)
    cross_sign = orientation.cross_sign(
        vacuum_parity, global_parity
    )
    regular[level] += (
        cross_sign * vacuum_coefficient * global[global_level]
    )
```

Here `orientation.cross_sign` evaluates the inversion sum (61); its implementation is `Code/ns_regular_block.py`. The lift monomial attached to this summand is  $\prod_e \xi_e^{v_e \bmod 2} \zeta_e^{r_e \bmod 2}$ , which equals the lift monomial at  $\ell = \mathbf{v} + \mathbf{r}$ .

For example, take

$$\ell = (0, 3, 4), \quad \mathbf{v} = (0, 3, 3), \quad \mathbf{r} = (0, 0, 1). \quad (210)$$

Their parity vectors give

$$\Omega_\Theta(\ell) = +1, \quad \Omega_\Theta(\mathbf{v}) = +1, \quad \Omega_\Theta(\mathbf{r}) = -1, \quad (211)$$

so Eq. (69) gives  $B_\Theta = 1$  and the cross sign is  $-1$ . At the benchmark point (217),

$$V_{(0,3,3)} = 1, \quad G_{(0,0,1)} = -0.427350427350427 \dots, \quad (212)$$

and this splitting contributes  $+0.427350427350427 \dots$ . The opposite sign from naive multiplication is therefore traced to one explicit half-edge inversion.

The left-hand side of the finite- $c$  check is constructed without any recursion. If  $\mathcal{B}_e(\ell_e)$  is the NS PBW basis and  $B_e^{(\ell_e)}$  its Gram matrix, the direct theta coefficient is

$$D_\ell(c) = \Omega_\Theta(\ell) \prod_e \xi_e^{\ell_e \bmod 2} \rho_{abc}(B_0^{-1})^{aa'} (B_1^{-1})^{bb'} (B_\infty^{-1})^{cc'} \rho_{a'b'c'}. \quad (213)$$

The code generates the three PBW bases, normal-orders the super-Virasoro algebra to form the Gram matrices, evaluates every entry of  $\rho$  from the NS Ward identities, and performs the displayed tensor contraction.

The regular input is computed independently. For the vacuum part, delete all PBW words containing  $G_{-1/2}$  or  $L_{-1}$ , sew the resulting irreducible vacuum bases by the same operation (213), and evaluate at five large generic central charges  $c_j$ . At every level the values are fitted as

$$D_v^{\text{vac}}(c_j) = V_v + \frac{a_1}{c_j} + \frac{a_2}{c_j^2} + \dots, \quad (214)$$

so the constant term gives the CCY-frame vacuum coefficient. This vacuum calculation has no generic internal weights and uses no Kac residues. It is then combined with the closed global coefficient by Eq. (63) to obtain  $U_\ell$ .

Finally, the program adds every admissible local Kac residue with  $rs \leq 2\ell_e$ . It lowers the twice-level on edge  $e$  by  $rs$ , shifts  $h_e$  by  $rs/2$ , toggles both adjacent vertex sectors when  $rs$  is odd, and applies the null-level lift and orientation-transport signs. The result is compared coefficient by coefficient with  $D_\ell(c)$ . At total level six the integer triples obey

$$n_0 + n_1 + n_\infty \leq 12, qquad n_e = 2\ell_e, \quad (215)$$

and hence there are

$$\#\{\mathbf{n}\} = \binom{12+3}{3} = 455 \quad (216)$$

independent coefficient comparisons.

Belavin–Geiko establish the sphere and torus versions explicitly [2]. Nothing in the counting uses a special graph: it applies independently at every trinion and every inverse metric.

## 6.6 Automated checks

The general recursion checks are implemented in

`Code/ns_genus_c_recursion_checks.py`

and are deliberately small and representation-theoretic. The script can be run from the repository root as

`python3 Code/ns_genus_c_recursion_checks.py`

or with `--json`. It performs the following independent checks.

1. It substitutes  $b_{3,1}(h)$  into  $h_{3,1}(c(b))$  and checks the pole equation.
2. It compares Eq. (26) with a centered finite difference of  $c_{3,1}(h)$ .
3. It evaluates Eq. (76) and the sphere coefficients (205), then compares them through twice-level eight with the production NS sphere seed in `Code/superconformal_blocks.py`.
4. It inverts Eq. (73) at  $c = 10^3$  and  $10^6$  and verifies suppression of the non-global state.
5. It constructs Eq. (109) by a product recursion and independently by explicit bosonic occupation/fermionic subset enumeration, for both lift signs.

The global calculation is implemented separately in

`Code/ns_global_osp_block.py`.

It implements the six independent kernels in Eq. (85), the two reflected kernels in Eq. (86), the three-chain completion (88), the parity projector, and the trifundamental coefficient (204). Its checks use Eq. (75), the endpoint reduction (203), the zero-level Ward table, the CCY middle-chain identity (89), and the two published torus coefficients (208).

The independent superspace derivation is implemented in

`Code/ns_osp_superspace.py`.

It first expands  $Z_{ij}$  and  $\eta_{123}$  in a symbolic three-generator exterior algebra and verifies that the result is exactly Eq. (116). It then differentiates that kernel according to Eq. (121) and compares against `Code/ns_global_osp_block.py`. The check covers all eight Grassmann sectors and all triples of occupations through three on two generic rational weight assignments; all 1024 descendant coefficients agree.

The genuine-supermoduli plumbing check is implemented in

`Code/ns_supermoduli_plumbing.py`.

It verifies  $\mathcal{D}^2 = \partial$ , checks that the even edge count and odd vertex count reproduce Eq. (133), and constructs the two-vertex sphere generating block explicitly. Ordered integration over its two vertex moduli reproduces 256 directly sewn sector coefficients through bosonic occupation three with zero error.

The independent exterior-algebra implementation

`Code/ns_grassmann_sewing.py`



constructs the edge kernel (124), performs the ordered Berezin integral, and compares it with the component contraction. It checks all eight theta-graph parity assignments, a punctured orientation with both external parities, and the even/odd inverse-Gram normalization (127).

The lifted Schottky calculation is implemented in

`Code/ns_vacuum_schottky.py`.

It enumerates primitive cyclic words modulo inversion, constructs the chosen  $SL(2)$  lift of each word, extracts  $(u_\gamma, k_\gamma)$ , and evaluates Eq. (107). Its tests are independent at the relevant steps:

1. compare  $u_\gamma^2$  with the projective multiplier for all primitive words through length five and verify cyclic/inverse invariance;
2. compare the genus-one Schottky evaluation with a direct oscillator product for both lift signs;
3. test the exterior-Fock trace-log identity (105) on a finite complex kernel;
4. compare the bosonic denominator exactly with the existing CCY implementation in `Code/python/genus2_vacuum_blocks.py`;
5. scale the theta plumbing parameters uniformly and verify convergence to the three-link coefficient (200) for all eight edge-lift assignments;
6. subtract the entire independently sewn vacuum table through total level six for all eight lift assignments. Under  $q_e \mapsto q_e/2$ , the maximum remainder ratio is 0.00647, consistent with a first omitted contribution at total level seven and incompatible with a mismatch at level six or below.

The half-edge sign calculation is implemented in

`Code/ns_regular_block.py`.

Given the ordered pairs of internal half-edges and the contraction permutation, it evaluates  $Q_\Gamma$  by inversion counting, checks that its linear BPZ part drops out of the polarization, and evaluates Eq. (63). For the theta graph it regenerates both Eqs. (196) and (69).

The direct finite- $c$  comparison is implemented in

`Code/ns_genus12_finite_c_check.py`.

It does not call the recursion to construct its left-hand side. It enumerates the NS PBW basis, normal-orders the super-Virasoro algebra to obtain every Gram matrix, evaluates the full three-descendant tensor from the NS Ward identities, inverts the three edge Gram matrices, and contracts the two theta-graph trinions. The genus-two cutoff is total plumbing level, the same convention as the repository's CCY bosonic checker. For the regular term it separately constructs the irreducible vacuum basis by deleting every PBW word containing  $G_{-1/2}$  or  $L_{-1}$ , computes the vacuum Gram matrices and Ward tensors, and takes the large- $c$  limit by a five-point Richardson extrapolation in  $1/c$ . It combines this table with the global coefficients using the explicit polarization routine, not a named product. Hence the vacuum table is not fitted from the generic finite- $c$  block used on the left-hand side.

For the reproducible generic point

$$c = 37.25, \quad (h_0, h_1, h_\infty) = (0.73, 0.91, 1.17), \quad (\xi_0, \xi_1, \xi_\infty) = (1, 1, 1), \quad (217)$$

the genus-one trace gives, through level 6,

$$1 + \xi q^{1/2} + q + 2\xi q^{3/2} + 3q^2 + 4\xi q^{5/2} + 5q^3 + 7\xi q^{7/2} + 10q^4 + 13\xi q^{9/2} + 16q^5 + 21\xi q^{11/2} + 28q^6. \quad (218)$$

This equals the generic NS character exactly: at each level the direct contraction is  $\text{Tr}(B^{-1}B)$ , so there is no residual  $c$ -dependence.

At genus two the script compares every triple  $(\ell_0, \ell_1, \ell_\infty)$  with  $\ell_0 + \ell_1 + \ell_\infty \leq 6$ : 455 coefficients in total. Some representative values are

| $2\ell$   | regular $U_\ell$   | direct finite $c$  | recursion          |
|-----------|--------------------|--------------------|--------------------|
| (3, 0, 0) | 0.261972379997773  | 0.263840424020828  | 0.263840424020828  |
| (0, 0, 3) | -0.437906238804443 | -0.438382932952886 | -0.438382932952887 |
| (1, 0, 4) | 0.108351800526099  | 0.175068774562786  | 0.175068774562786  |
| (3, 3, 0) | -1.00317218140405  | -1.04112826090143  | -1.04112826090143  |
| (3, 0, 3) | 1.24049898969217   | 1.12019936770983   | 1.12019936770983   |
| (0, 3, 3) | 3.81585349350960   | 3.95674683480596   | 3.95674683480596   |
| (0, 4, 2) | 4.70247533416103   | 4.87540464672133   | 4.87540464672133   |
| (0, 0, 6) | 0.554995640172376  | 0.618077510729804  | 0.618077510729803  |

The maximum direct-versus-recursive error at each total twice-level  $0, 1, \dots, 12$  is

$$\begin{aligned} &(0, 0, 1.39 \times 10^{-16}, 2.22 \times 10^{-16}, 3.33 \times 10^{-16}, \\ &7.77 \times 10^{-16}, 8.88 \times 10^{-16}, 2.66 \times 10^{-15}, 7.99 \times 10^{-15}, \\ &1.42 \times 10^{-14}, 2.84 \times 10^{-14}, 4.26 \times 10^{-14}, 5.36 \times 10^{-13}). \end{aligned} \quad (219)$$

The worst coefficient is  $2\ell = (0, 1, 11)$ . Two additional generic benchmark points give maximum errors  $1.28 \times 10^{-12}$  and  $8.70 \times 10^{-14}$ . The independent vacuum-quotient extraction agrees with the integer table (202) to  $1.24 \times 10^{-14}$ .

This higher cutoff is also a falsification test. Keeping the correct vacuum coefficients but deleting the polarized sign in Eq. (182) gives an error 1.37 already at  $2\ell = (4, 0, 3)$ . Thus the level-three pass could not have detected the missing cross sign, because the first vacuum terms there only multiply the constant global coefficient.

The current output of all five scripts includes

| Check                                                                                                   | Result                         |
|---------------------------------------------------------------------------------------------------------|--------------------------------|
| Kac pole equation error                                                                                 | $3.33 \times 10^{-16}$         |
| Jacobian relative error                                                                                 | $1.50 \times 10^{-10}$         |
| Maximum sphere global-seed error                                                                        | 0                              |
| Level-3/2 non-global error at $c = 10^3$                                                                | $1.14 \times 10^{-3}$          |
| Level-3/2 non-global error at $c = 10^6$                                                                | $1.14 \times 10^{-6}$          |
| Vacuum product versus Fock enumeration                                                                  | pass                           |
| Global norm recursion maximum error                                                                     | $7.11 \times 10^{-15}$         |
| Endpoint and primary Ward-table errors                                                                  | 0                              |
| CCY middle-chain identity error                                                                         | $3.55 \times 10^{-14}$         |
| Torus levels 1/2 and 1 errors                                                                           | 0, $3.33 \times 10^{-16}$      |
| Trinion parity projector                                                                                | pass                           |
| Lift relation $u_\gamma^2 = k_\gamma$ , maximum relative error                                          | $9.95 \times 10^{-16}$         |
| Cyclic/inverse half-multiplier error                                                                    | $4.58 \times 10^{-16}$         |
| Lifted genus-one product error                                                                          | $2.22 \times 10^{-16}$         |
| Exterior trace-log error                                                                                | 0                              |
| Bosonic reduction versus CCY                                                                            | 0                              |
| Theta leading-error ratio under $t \mapsto t/2$                                                         | 0.489                          |
| Corrected primitive product versus the level-six vacuum table:<br>remainder ratio under $t \mapsto t/2$ | $0.00647 < 2^{-7}$             |
| General half-edge polarization algorithm                                                                | pass                           |
| Vacuum quotient versus level-six seed                                                                   | $1.24 \times 10^{-14}$ maximum |
| Genus-one direct versus recursion through level 6                                                       | exact                          |
| Genus-two direct versus recursion, 455 coefficients                                                     | $5.36 \times 10^{-13}$ maximum |

These checks certify the pole data, the global factor, the direct CCY-frame vacuum seed, and the assembled finite- $c$  recursion through total level six. The corrected product remainder scales as level seven after subtracting the entire level-six direct table, so it excludes a product/sewing mismatch at any displayed level. The all-order identification itself follows from the primitive-link derivation, rather than from this finite truncation.

## 6.7 Direct finite- $c$ stress tests

The preceding checks test the ingredients of the construction separately. We also performed a direct coefficient-by-coefficient comparison of the assembled recursion with the finite-level block. This is the cleanest test of the recursion formula itself. Write

$$D_{\mathbf{n}}(c, \mathbf{h}) \quad \text{and} \quad R_{\mathbf{n}}(c, \mathbf{h}), \quad \mathbf{n} = (n_0, n_1, n_\infty) = 2\boldsymbol{\ell}, \quad (220)$$

for, respectively, the coefficient obtained by direct PBW sewing and the coefficient obtained by expanding the recursion relation to the same twice-level. No numerical power-series extrapolation is involved: at a fixed cutoff both constructions produce the coefficient itself. Possible floating-point error enters only if the resulting rational functions are subsequently evaluated numerically.

### 6.7.1 Exact symbolic comparison through total order three

The script `Code/ns_genus2_symbolic_low_order.py` carries out the two constructions independently over  $\mathbb{Q}(c, h_0, h_1, h_\infty)$ . Direct Gram-matrix inversion and Ward contractions are used on one

side, whereas the global seed, Kac residues, weight shifts, and edge-orientation signs are assembled on the other. For all

$$n_0 + n_1 + n_\infty \leq 6, \quad (221)$$

the script simplifies  $D_{\mathbf{n}} - R_{\mathbf{n}}$  as an exact rational function. Thus this range contains all  $84 = \binom{9}{3}$  coefficients through total physical order three, rather than a sample of coefficients.

The relevant degenerate central charges in this range are

$$\begin{aligned} c_{3,1}(h) &= 6 - 3h - \frac{6}{2h+1} = -\frac{3h(2h-3)}{2h+1}, \\ c_{2,2}(h) &= \frac{3}{2} - 8h, \\ c_{5,1}(h) &= \frac{13}{2} - h - \frac{9}{h+1} = -\frac{(h-5)(2h-1)}{2(h+1)}. \end{aligned} \quad (222)$$

Representative identities expose more than agreement of two final numbers. Define

$$\Delta_0 = (2h_0 + 1)c + 6h_0^2 - 9h_0 = (2h_0 + 1)(c - c_{3,1}(h_0)). \quad (223)$$

At twice-level  $(3, 0, 0)$ , direct sewing gives

$$\begin{aligned} U_{300} &= \frac{(2h_0 + 2h_1 - 2h_\infty + 1)^2}{8h_0(2h_0 + 1)}, \\ D_{300} - U_{300} &= \frac{6(h_1 - h_\infty)^2}{(2h_0 + 1)\Delta_0} = \frac{1}{c - c_{3,1}(h_0)} \frac{6(h_1 - h_\infty)^2}{(2h_0 + 1)^2}. \end{aligned} \quad (224)$$

The first term is the global coefficient and the second is exactly the  $(3, 1)$  odd residue prescribed by the recursion.

The mixed coefficient  $(3, 1, 0)$  checks both the shifted block and the orientation sign. With

$$\mathcal{A} = 2h_0h_1 + 2h_0h_\infty - h_0 - 2h_1^2 + 4h_1h_\infty + h_1 - 2h_\infty^2 + h_\infty, \quad (225)$$

one finds

$$D_{310} - U_{310} = -\frac{3\mathcal{A}^2}{4h_1(2h_0 + 1)\Delta_0} = -\frac{1}{c - c_{3,1}(h_0)} \frac{3\mathcal{A}^2}{2(2h_0 + 1)^2} \frac{1}{2h_1}. \quad (226)$$

Here  $1/(2h_1)$  is the shifted global coefficient and the minus sign is the theta-graph orientation transport. At twice-level  $(4, 0, 0)$ , the same comparison resolves the two distinct pole channels separately:

$$D_{400} = U_{400} + \frac{R_{3,1}^{\text{even}}(h_0)}{c - c_{3,1}(h_0)} \frac{1}{2h_0 + 3} + \frac{R_{2,2}^{\text{even}}(h_0)}{c - c_{2,2}(h_0)}, \quad (227)$$

where

$$\begin{aligned} R_{3,1}^{\text{even}}(h_0) &= \frac{3\mathcal{A}^2}{2(2h_0 + 1)^2}, \\ R_{2,2}^{\text{even}}(h_0) &= \frac{\mathcal{B}^2}{4h_0(2h_0 + 3)}, \\ \mathcal{B} &= h_0^2 + 2h_0h_1 + 2h_0h_\infty - 3h_1^2 + 6h_1h_\infty - 3h_\infty^2. \end{aligned} \quad (228)$$

The identities at  $(5, 0, 0)$ ,  $(3, 3, 0)$ , and  $(6, 0, 0)$  additionally exercise the  $(5, 1)$  channel, simultaneous edge contributions, and all channels available at the cutoff; each simplifies identically to zero. Altogether the exact result is

$$D_{\mathbf{n}} - R_{\mathbf{n}} = 0 \quad \text{for all } \mathbf{n} \in \mathbb{Z}_{\geq 0}^3, \quad |\mathbf{n}| \leq 6, \quad 84/84 \text{ identities.} \quad (229)$$

The same calculation fixes the first nontrivial theta-graph vacuum seed directly from the large- $c$  limit:

$$\lim_{c \rightarrow \infty} (D_{033} - G_{033}) = 1, \quad \lim_{c \rightarrow \infty} (D_{303} - G_{303}) = 1, \quad \lim_{c \rightarrow \infty} (D_{330} - G_{330}) = -1. \quad (230)$$

This verifies the relative sign in the first theta contribution without using the Schottky product as input. As a secondary check, evaluating all 84 symbolic answers at an independent generic point agrees with the direct and recursive numerical implementations to  $1.78 \times 10^{-15}$  and  $3.11 \times 10^{-15}$ , respectively.

### 6.7.2 Finite-level stress test through total order eight

The independent finite-level implementation `Code/ns_genus12_finite_c_check.py` was then used at larger cutoffs. First, the previously reported order-six test was rerun without modifying its conventions or benchmark. All  $455 = \binom{15}{3}$  coefficients with  $n_0 + n_1 + n_\infty \leq 12$  agree, with maximum absolute discrepancy  $5.36 \times 10^{-13}$ . The earlier order-six check is therefore faithful.

For the order-eight test we chose the generic point

$$c = 41.3, \quad (h_0, h_1, h_\infty) = (0.731, 0.913, 1.173), \quad (\xi_0, \xi_1, \xi_\infty) = (1, 1, 1). \quad (231)$$

All  $969 = \binom{19}{3}$  coefficients with  $n_0 + n_1 + n_\infty \leq 16$  pass. The largest absolute discrepancy is  $1.071 \times 10^{-9}$ , at  $(0, 5, 11)$ , where the two values are

$$767.8733692113085 \quad \text{and} \quad 767.8733692123799. \quad (232)$$

After division by the natural coefficient scale, the largest discrepancy over the full table is  $9.124 \times 10^{-11}$ . These residuals are solely from floating-point evaluation of Gram inverses, residues, and shifted blocks; the comparison is between coefficients at a fixed cutoff and hence has no truncation remainder.

Two vacuum checks accompany the finite- $c$  comparison. Direct large- $c$  extraction through order eight produces 97 nonzero integer vacuum-seed coefficients, with maximum numerical extraction error  $2.84 \times 10^{-13}$ . Subtracting the complete order-eight table from the corrected primitive Schottky product gives a remainder ratio

$$\frac{|\mathcal{E}(t/2)|}{|\mathcal{E}(t)|} = 0.002881861 \quad (t = 0.08), \quad (233)$$

uniformly for all eight choices of the three edge lifts. This is compatible with the first omitted terms beginning at total order nine, at the finite values of  $t$  used in the test.

The higher-order run also exposed a useful exceptional-point issue. The old order-six benchmark remains generic throughout its tested range, but at higher shifted levels it encounters a coincident pole: at top twice-level  $(10, 3, 0)$  and  $c = 1.1423404255\dots$ , the shifted  $(5, 1)$  pole on edge zero coincides with the shifted  $(3, 1)$  pole on edge one. The formal recursion then requires weight detuning or a higher-order Laurent recursion; the scalar numerical evaluator now reports this unsupported collision explicitly. The exact double pole in Eq. (51) shows why merging equal simple-pole keys would not suffice. This is not a disagreement of generic coefficients, but it explains why a benchmark that is safe through order six need not remain safe at order eight.

In summary, on the generic locus the recursion has passed an exact symbolic comparison of every coefficient through total order three and an independent numerical stress test of every coefficient through total order eight. These tests strongly support the order-by-order prescription, including

its residue channels, shifted sub-blocks, lift signs, and vacuum seed in the theta-graph orientation convention. The sign compiler above reconstructs that convention from its half-edge frames and transition lifts, including  $(\beta_0, \beta_1, \beta_\infty) = (0, 0, 1)$ . The tests do not implement exceptional higher-order poles.

## 6.8 Companion code

The formulas are implemented in a small set of independent scripts:

| Script                                         | Role                                                                                                  |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Code/superconformal_blocks.py                  | sphere NS $c$ -recursion kernel                                                                       |
| Code/superconformal_torus_blocks.py            | toric NS $h$ -recursion and spin-lifted evaluation                                                    |
| Code/ns_global_osp_block.py                    | component global module and trinion kernels                                                           |
| Code/ns_regular_block.py                       | endpoint reflection/Koszul signs, frame-derived orientation, polarization, and regular-block assembly |
| Code/ns_vacuum_schottky.py                     | lifted primitive Schottky product                                                                     |
| Code/compare_ns_sphere_c_h_recursion.py        | order-ten sphere $c$ - versus $h$ -recursion comparison at $\hat{c} = 9$                              |
| Code/stress_ns_crossing.py                     | direct-recursion nonchiral sphere crossing test at $\hat{c} = 9$                                      |
| Code/compare_ns_torus_c_h_recursion.py         | order-ten torus one-point $c$ - versus $h$ -recursion comparison at $\hat{c} = 9$                     |
| Code/stress_ns_torus_modularity_c_recursion.py | direct functional- $c$ NS–NS torus modularity test                                                    |
| Code/ns_genus12_finite_c_check.py              | direct genus-one and genus-two finite- $c$ sewing                                                     |
| Code/ns_genus2_symbolic_low_order.py           | exact symbolic coefficient comparison                                                                 |
| Code/ns_osp_superspace.py,                     | optional superspace and supermoduli reformulations                                                    |
| Code/ns_supermoduli_plumbing.py                |                                                                                                       |

## Conclusion

The NS  $c$ -recursion has two independent ingredients. Representation theory fixes every moving pole and its edge-local residue. The CCY large- $c$  separation fixes the regular term as the graded assembly of the explicit global  $\mathfrak{osp}(1|2)$  network and the universal lifted vacuum product. Together they give the functional recursion (2) and its triangular coefficient form (50) at generic weights for an arbitrary ordered plumbing graph with primary or global external states, using Eq. (45) whenever an edge closes on one trinion. Equations (37) and (59) compute the endpoint and BPZ/spin-frame signs from the ordered lifted local coordinates, so no independent sign table is left unspecified. The present manuscript does not extend the recursion to coincident Kac loci or to non-global external descendants without additional large- $c$  external factors.

The genus- $g$ , zero-point block is the literal  $N = 0$  specialization (92); in the irreducible vacuum channel it reduces to the primitive product (93). Direct PBW sewing confirms the assembled recursion at genus one and genus two through the cutoffs recorded in Section 6. Superspace provides a compact reformulation, but no step of the main derivation depends on the exploratory odd-moduli plumbing proposal.

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